Abstract

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Chapter 10: Parametric Equations and Polar Coordinates

10.1: Curves Defined by Parametric Equations

Parametrization of a curve means defining x and y in terms of a third variable t called a **parameter**. This creates **parametric equations**:

$$x = f(t) \qquad y = g(t)$$

Each value of t determines a point (x, y). As t varies over its domain, (x, y) traces a curve C called a **parametric curve**. (Note that t does not necessarily represent time, and we could use other variable names instead of t).

Info

It is entirely possible for two distinct sets of parametric equations to generate the same set of points. It is also possible for two distinct values of t to generate the same (x, y).

For instance, $x = \cos t, y = \sin t$ and $x = \sin t, y = \cos t$ generate the same set of points (a circle). However, the first set of parametric equations traces the curve in the counterclockwise direction, while the second set traces the curve in the clockwise direction.

Example

Describe the parametric curve parametrized by the following: (h, k, r are constants)

$$x = h + r\cos t \qquad y = k + r\sin t$$

We can attempt to eliminate the parameter t to find an equation in terms of x, y.

$$\begin{gathered} r\cos t = x - h \qquad r\sin t = y - k \\ r^2\cos^2 t + r^2\sin^2 t = (x-h)^2 + (y-k)^2 \\ r^2 = (x-h)^2 + (y-k)^2 \end{gathered}$$

Thus, this set of parametric equations describes a circle centered at (h, k) with radius r.

With parameterization, we can also *restrict* t to a certain interval, i.e. $a \le t \le b$. For a curve parametrized by the following conditions,

$$x = f(t) \qquad y = g(t) \qquad a \le t \le b$$

it has **initial point** $(f(a), g(a))$ and **terminal point** $(f(b), g(b))$.

Exercise:
Eliminate the parameter t for the equations

$$x = t^2 + t \qquad y = t^2 - t$$

$$x - y = 2t \qquad x + y = 2t^2$$
$$(x-y)^2 = 4t^2 = 2(x+y)$$
$$x^2 + y^2 + 2xy - 2x - 2y = 0$$

This becomes a parabola aligned along $x = y$.

Example: Cycloid

The cycloid is a curve traced by a point P on the circumference of a circle that rolls along a straight line. (Page 683, Figure 13).

We will skip the derviation of the parametrization of the cycloid. (See Page 684 if interested). It is described by: (r is constant)

$$x = r(\theta - \sin\theta) \qquad y = r(1 - \cos\theta) \qquad \theta \in \mathbb{R}$$

Let's now eliminate the parameter θ

$$r\theta - x = r\sin\theta \qquad r - y = r\cos\theta$$
$$r^2\sin^2\theta + r^2\cos^2\theta = (r\theta - x)^2 + (r-y)^2$$
$$r^2 = (r\theta - x)^2 + (r-y)^2$$
$$\sqrt{r^2 - (r-y)^2} = r\theta - x$$
$$\theta = \frac{\sqrt{r^2 - (r-y)^2} + x}{r}$$

Having solved for θ in terms of x, y, r, we can effectively substitute for θ in the original parametric equations to eliminate the parameter. (Although, it is a very, very ugly equation).

10.2 Calculus with Parametric Curves

Info

In this section, "Cartesian Curve" will refer to a curve C represented by equations with variables x, y, and "Parametric Curve" will refer to a curve C represented by equations $x(t), y(t)$.

Derivative

Deriving $\frac{dy}{dx}$

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$
$$\text{If } \frac{dx}{dt} \neq 0: \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}.$$

Deriving $\frac{d^2y}{dx^2}$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$
$$= \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

The last step above is simply expanding $\frac{d}{dx}$ into $\frac{\frac{d}{dt}}{\frac{dx}{dt}}$.

Formulas

> **First and Second Derivative of a Parametric Curve**
>
> $$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$
> $$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

Horizontal and Vertical Tangents

The tangent line is...

- horizontal when $\frac{dy}{dx} = 0 \implies \frac{dy}{dt} = 0$.
- vertical when $\frac{dy}{dx} = \text{undefined} \implies \frac{dx}{dt} = 0$.
- unknown when $\frac{dx}{dt} = \frac{dy}{dt} = 0$; L'Hôpital's Rule is needed.

Area

Suppose that some curve C is traced by $x = f(t)$ and $y = f(t)$ as the parameter increases from $\alpha \to \beta$. Then, if $a = f(\alpha)$ and $b = f(\beta)$:

$$\begin{aligned} A &= \int_a^b y\,dx \\ &= \int_\alpha^\beta y\,\frac{dx}{dt}dt \end{aligned}$$

Note that, if instead $b = f(\alpha)$ and $a = f(\beta)$, we can simply reverse the interval for the integral (the one in respect to t).

> **Area under a Parametric Curve**
>
> $$A = \int_\alpha^\beta y\frac{dx}{dt}\,dt$$

Arc Length

Deriving for a Cartesian Curve

Let curve C be described by $y = F(x)$, where F is differentiable. Additionally, let $x = f(t)$ and $y = g(t)$. Then the arc length L of C between $a \le x \le b$ can be obtained by partitioning $[a, b]$ into n subintervals of equal length, deriving an approximation for L based on the partitions, and solving for $\lim_{n\to\infty} L$ to derive the arc length formula.

First, the partitioning:

$$a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b$$

The length of each subinterval is $\Delta x = \frac{b-a}{n}$.

For each subinterval $x_{i-1} \to x_i$, we draw a line between the points $(x_{i-1}, F(x_{i-1}))$ and $(x_i, F(x_i))$. Let Δs_i be the length of this line. Additionally, let $\Delta x_i = x_i - x_{i-1}$ (yes, this is equivalent to Δx) and let $\Delta y_i = y_i - y_{i-1}$. Then:

$$\Delta s_i = \sqrt{(\Delta x_i)^2 + (\Delta y_i^2)}$$

Now, the approximation of L is simply:

$$L \approx \sum_{i=1}^{n} \Delta s_i$$

Taking the limit:

$$L = \lim_{n\to\infty} \sum_{i=1}^{n} \Delta s_i$$

Remember that $\Delta s_i = ds$. Rewritten, we can express this as

$$L = \int_a^b ds$$

It suffices to then find an expression for ds a.k.a. Δs_i.

$$\begin{aligned}\Delta s_i &= \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2} \\ &= \sqrt{1 + \left(\frac{\Delta x_i}{\Delta y_i}\right)^2}\Delta x_i \\ &= \sqrt{1 + \left(\frac{\Delta x_i}{\Delta y_i}\right)^2}\Delta x\end{aligned}$$

In other words,

$$ds = \sqrt{1 + \left(\frac{\Delta x_i}{\Delta y_i}\right)^2} dx$$

And thus,

$$L = \int_a^b \sqrt{1 + \left(\frac{\Delta x_i}{\Delta y_i}\right)^2} dx$$

And we're done!

Formal Interlude

By the **Mean Value Theorem** (MVT), since $y = F(x)$ is differentiable, $\forall i \in \{1, \ \dots, \ n\}, \exists x_i^* \in (x_{i-1}, x_i)$ such that

$$F'(x_i^*) = \frac{\Delta y_i}{\Delta x_i}$$

i.e. there is some x_i^* in the interval with the same slope as the slope of the line segment between the points $(x_{i-1}, F(x_{i-1}))$ and $(x_i, F(x_i))$.

Then,

$$\begin{aligned}L &= \lim_{n\to\infty} \Delta s_i \\ &= \lim_{n\to\infty} \sqrt{1 + \left(\frac{\Delta y_i}{\Delta x_i}\right)^2}\Delta x \\ &= \lim_{n\to\infty} \sqrt{1 + [F'(x_i)]^2 \Delta x} \\ &= \int_a^b \sqrt{1 + [F'(x_i)]^2}\, dx\end{aligned}$$

This is a slightly more formal last step to the proof, but it returns the same answer.

ⓘ Arc Length for a Cartesian Curve

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

Deriving for a Parametric Curve

The derivation for the arc length formula for a parametric curve follows simply from the formula for a Cartesian curve.

$$\begin{aligned} L &= \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \\ &= \int_\alpha^\beta \sqrt{1 + \left(\frac{\frac{dy}{dt}}{\frac{dx}{dt}}\right)^2}\, \frac{dx}{dt} dt \\ &= \int_\alpha^\beta \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \end{aligned}$$

Arc Length for a Parametric Curve

$$L = \int_\alpha^\beta \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

Surface Area

Deriving for a Cartesian Curve

We can revolve a curve C around an axis to obtain a 3-dimensional shape C'. For our purposes, we will consider rotating C around the x-axis on the interval $a \le x \le b$ for our derivation.

Just like in the 2-dimensional derivation for arc length, we will approximate C by partitioning it into n subintervals of equal length. Then, by rotating the line segments formed by the subintervals, we obtain an approximation for C'.

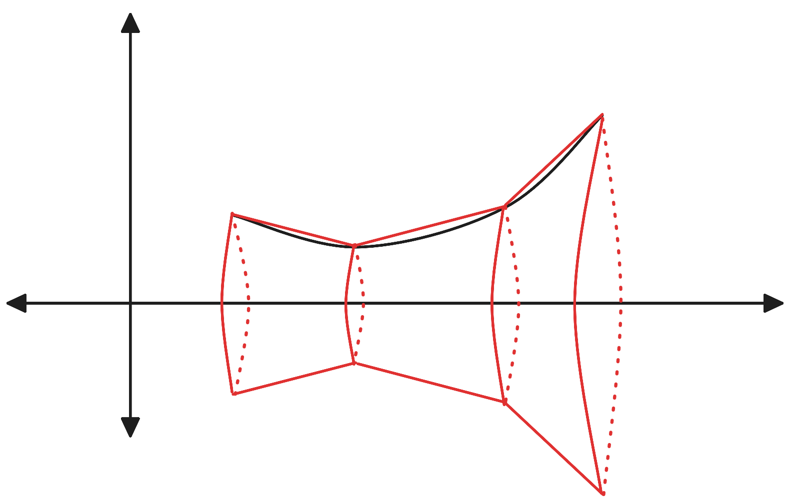
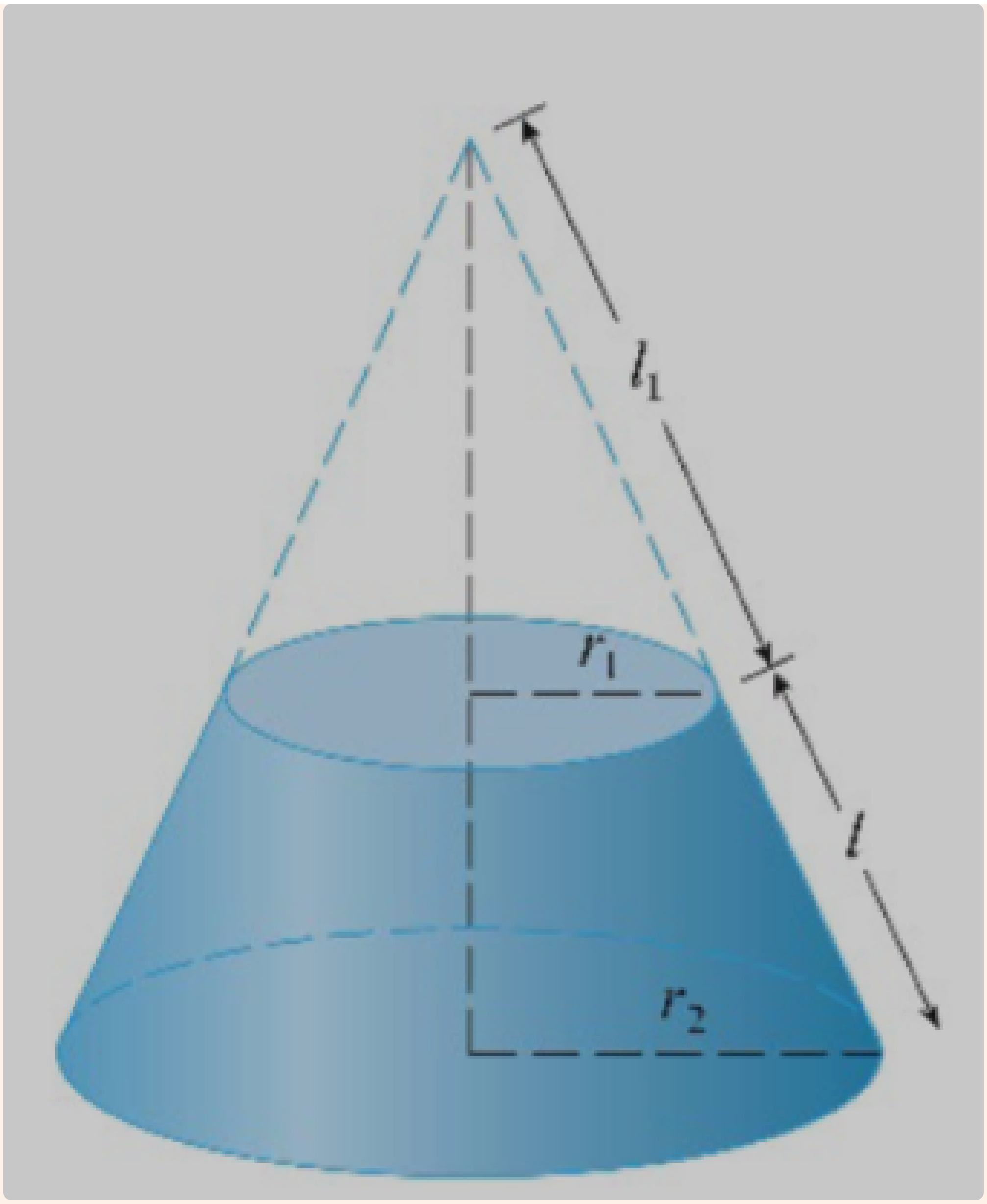


Figure 1. A diagram illustrating the method of approximating the shape of revolution of a Cartesian curve.

We can calculate the surface area of this shape by computing the lateral surface area of a frustum. *Note that the lateral surface area denotes the surface area of the frustum that does not include the circular bases.

Frustum?

A frustum is created when a cross section parallel to the cone's base (circle) splits the cone into two shapes: a cone above and a frustum below.



The lateral surface area of a frustum is

$$A = 2\pi r l$$

Where $r = \frac{1}{2}(r_1 + r_2)$.
**The proof of this will be left out for brevity. Search it up if you're interested!*

Thus, we can approximate the surface area of the shape of revolution by summing the surface areas of the frustums. Note that the symbols used here are equivalent to the symbols used here.

$$S \approx \sum_{i=1}^{n} A_i = \sum_{i=1}^{n} 2\pi \frac{y_{i-1} + y_i}{2} \Delta s_i$$

Δs_i is the same as before. So we can write:

$$
\begin{aligned}
S &= \lim_{n\to\infty}\sum_{i=1}^{n} 2\pi\frac{y_{i-1}+y_i}{2}\sqrt{1+\left(\frac{\Delta x_i}{\Delta y_i}\right)^2}\Delta x \\
&= \int_a^b 2\pi y\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,dx
\end{aligned}
$$

** Note that y is substituted for $\frac{y_{i-1}+y_i}{2}$ for y because y_{i-1} and y_i become infinitely close to each other as n tends to ∞. y can be substituted with $F(x)$ when actually solving.

> **Surface Area of a Cartesian Curve**
>
> $$S = \int_a^b 2\pi y\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,dx$$

Deriving for a Parametric Curve

The process here is identical to the parametric derivation for arc length.

$$
\begin{aligned}
S &= \int_a^b 2\pi y\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,dx \\
&= \int_\alpha^\beta 2\pi y\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\frac{\mathrm{d}x}{\mathrm{d}t}dt \\
&= \int_\alpha^\beta 2\pi y\sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2+\left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2}\,dt
\end{aligned}
$$

> **Surface Area of a Parametric Curve**
>
> $$S = \int_\alpha^\beta 2\pi y\sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2+\left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2}\,dt$$

Generalization

You may not necessarily rotate a curve around the x-axis. Generally, you will be assigned to rotate around some line $y = k$ (vertical) or $x = h$ (horizontal). The formula is always similar, however. You simply replace y with a formula for the radius r.

Reference Formulas

Formulas

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \qquad \frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

$$A = \int_\alpha^\beta y\frac{dx}{dt}\,dt$$

$$L = \int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx \qquad L = \int_\alpha^\beta \sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt$$

$$S = \int_a^b 2\pi y\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,dx \qquad S = \int_\alpha^\beta 2\pi y\sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2+\left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2}\,dt$$

10.3 Polar Coordinates

Intro

The polar coordinate system is an alternative to the Cartesian coordinate system, where a point P is represented by (r, θ). r is the length of the line segment between the **pole** O, which is typically the origin, and the point P. (We choose the variable r because it means *radius*). θ is the angle between the line segment and the **polar axis**. The polar axis is a ray that is typically equivalent to the positive x-axis.

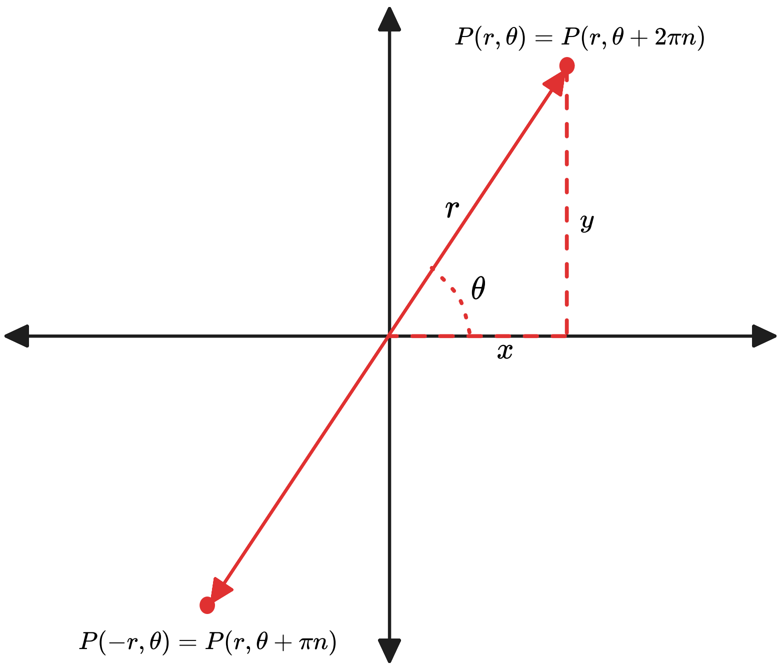


Figure 2. A graph demonstrating polar coordinates. on the Cartesian plane.

Note that $P(-r,\theta) = P(r,\theta+\pi n)$ for $n \in \mathbb{Z}$ and n **odd**. Also note that $P(r,\theta) = P(r,\theta+2\pi n)$ for $n \in \mathbb{Z}$.

Polar curves are essentially just a special form of parametric curves. Many of the same ideas/formulas for parametric curves will apply here too!

Cartesian to Polar

We can write a couple equations that represent the relationship between polar and Cartesian coordinates.

Based on the definitions of $\cos\theta$ and $\sin\theta$, we can write

$$\cos\theta = \frac{x}{r} \qquad \sin\theta = \frac{y}{r}$$

It may help to refer to **Figure 2** to see why these equations are true.

From this, it's easy to derive the following:

Polar $\rightarrow$ Cartesian Equations

$$x = r\cos\theta \qquad y = r\sin\theta$$

Subsequently, we may derive the following:

Cartesian $\rightarrow$ Polar Equations

$$r^2 = x^2 + y^2 \qquad \tan\theta = \frac{y}{x}$$

Finally, note that the equation of a polar curve is typically denoted as $r = f(\theta)$.

Symmetry

When sketching polar curves, recognizing symmetry can help significantly.

- $f(\theta) = f(-\theta) \implies$
  The curve is symmetric about the polar axis.
- $f(\theta) = -f(\theta) \iff f(\theta) = f(\theta + \pi) \implies$
  The curve is symmetric about the pole.
- $f(\theta) = f(\pi - \theta) \implies$
  The curve is symmetric about $\theta = \frac{\pi}{2}$ (equivalent to $y = x$)

Derivatives

To derive polar curves, we essentially consider them as parametric. (Because, that basically is what they are!)

As a reminder,

$$x = r\cos\theta \qquad y = r\sin\theta$$

Deriving,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta \qquad \frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta$$

Therefore,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\frac{\mathrm{d}y}{\mathrm{d}\theta}}{\frac{\mathrm{d}x}{\mathrm{d}\theta}} = \frac{\frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta}{\frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta}$$

Derivative for a Polar Curve

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta}{\frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta}$$

For some $\theta = k$:

- $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0 \wedge \frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0 \implies$ Vertical Tangent.
- $\frac{\mathrm{d}y}{\mathrm{d}\theta} \neq 0 \wedge \frac{\mathrm{d}y}{\mathrm{d}\theta} = 0 \implies$ Horizontal Tangent.
- $\frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{\mathrm{d}x}{\mathrm{d}\theta} = 0 \implies$ Use L'Hôpital's Rule to calculate $\lim_{\theta\to k}\frac{\mathrm{d}y}{\mathrm{d}x}$.

Derivative at $f(\theta) = r = 0$

(Equivalent to a derivative at the pole)

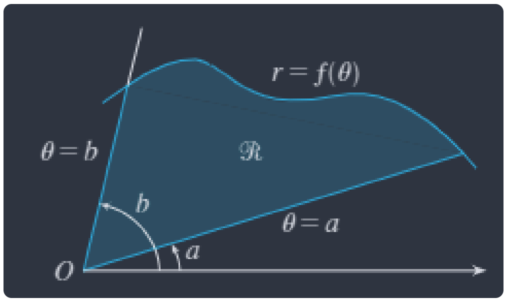
$$\frac{\mathrm{d}y}{\mathrm{d}x} = \tan\theta \qquad \text{if } \frac{\mathrm{d}r}{\mathrm{d}\theta} \neq 0$$

To find the 2nd derivative, we can follow the same steps as for a parametric curve. (Omitted for brevity).

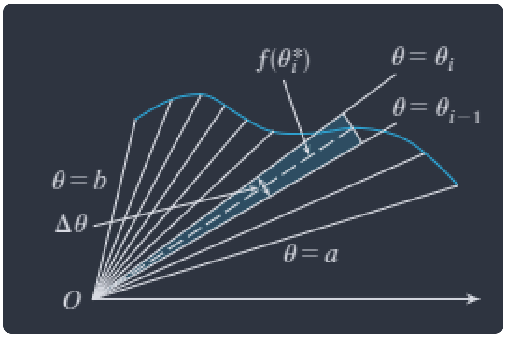
10.4 Areas and Lengths in Polar Coordinates

Area

We define the area of a polar curve as the area swept out by the radius $r = f(\theta)$ between bounds a and b on θ. For instance,



We can approximate this area by summing several sectors of circles:



The area of a sector of a circle is

$$A = \frac{1}{2}r^2\theta$$

Therefore, we can approximate the area of the polar curve as

$$\Delta A_i = \sum_{i=1}^{n} \frac{1}{2}[f(\theta_i)]^2 \Delta\theta$$

We take the limit as $\Delta\theta \to \infty \implies n \to \infty$:

$$\lim_{n\to\infty} \sum_{i=1}^{n} \frac{1}{2}[f(\theta_i)]^2 \Delta\theta = \int_a^b \frac{1}{2}[f(\theta)]^2 \, d\theta$$

Therefore,

Area of a Polar Curve

$$A = \int_a^b \frac{1}{2}r^2 \, d\theta$$

Arc Length

For the arc length of a polar curve, it suffices to repurpose the equation for the arc length of a parametric curve. Remember:

$$L = \int_a^b \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}\theta}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}\theta}\right)^2} \, d\theta$$

By definition,

$$x = r\cos\theta \qquad y = r\sin\theta$$

Differentiating,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta \qquad \frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta$$

So,

$$\begin{aligned}\left(\frac{\mathrm{d}x}{\mathrm{d}\theta}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}\theta}\right)^2 &= \left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2 \cos^2\theta - 2r\frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta\sin\theta + r^2 + \sin^2\theta + \left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2 \sin^2\theta + 2r\frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta\cos\theta + r^2\cos^2\theta \\ &= \left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2 + r^2\end{aligned}$$

Thus,

Arc Length of a Polar Curve

$$L = \int_a^b \sqrt{r^2 + \left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2}\, d\theta$$

Chapter 12: Vectors and the Geometry of Space

Chapter omitted because this is more of a linear algebra basics chapter. But, here's a nice reference sheet:

References

Dot Product

$$\begin{aligned} u \cdot v &= |u||v|\cos\theta \\ u \cdot v &= 0 \implies u \perp v \\ \\ u \cdot u &= |u|^2 \\ u \cdot v &= v \cdot u \\ u \cdot (v + w) &= u \cdot v + u \cdot w \\ (\alpha u) \cdot v &= \alpha(u \cdot v) = u \cdot (\alpha v) \\ \mathbf{0} \cdot u &= 0 \end{aligned}$$

Projections

$$\begin{aligned} \mathrm{comp}_u v &= \frac{u \cdot v}{|u|} \\ \mathrm{proj}_u v &= \left(\frac{u \cdot v}{|u|}\right)\frac{u}{|u|} \end{aligned}$$

Cross Product

$$\begin{aligned} (u \times v) &\perp u, v \\ |u \times v| &= |u||v|\sin\theta \\ u \times v &= \mathbf{0} \implies u \perp v \\ |u \times v| &= \text{area of parallelogram determined by } u \text{ and } v \\ i \times j = k \qquad & j \times k = i \qquad k \times i = j \\ j \times i = -k \qquad & k \times j = -i \qquad i \times k = -j \\ u \times v &= -u \times v \\ (\alpha u) \times b &= \alpha(u \times v) = u \times (\alpha b) \\ u \times (v + w) &= u \times v + u \times w \\ (u + v) \times w &= u \times w + v \times w \\ u \cdot (v \times w) &= (u \times v) \cdot w \\ u \times (v \times w) &= (u \cdot w)v - (u \cdot v)w \end{aligned}$$

Triple Product

$$V = |u \cdot (v \times w)|$$

(V is the volume of the parallelepiped determined by vectors u, v, w)
Distances

$$\text{Distance from point to line}: \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$$
$$\text{Distance from point to plane}: \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$$
$$\text{Distance between 2 parallel planes}: \frac{|d_2 - d_1|}{\sqrt{a^2 + b^2 + c^2}}$$

Chapter 13: Vector Functions

13.1 Vector Functions and Space Curves

A **vector-valued function** is a function whose domain is a set of real numbers, and whose range is a set of vectors. For instance,

$$r(t) = \langle f(t), g(t), h(t) \rangle = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

Where $r(t)$ is the vector-valued function and f, g, h are the **component functions** of r. Often (for 3-dimensions), $r : \mathbb{R} \to \mathbb{R}^3$.

Limits and Continuity

The limit for a vector-valued function is simply:

$$\lim_{t \to a} r(t) = \langle \lim_{t \to a} f(t), \lim_{t \to a} g(t), \lim_{t \to a} h(t) \rangle$$

And $r(t)$ is continuous at $t = a$ when

$$\lim_{t \to a} r(t) = r(a)$$

Space Curves

Suppose f, g, t are continuous, real-valued functions defined on an interval ℓ. Then the set of C of all points (x, y, z) in space described by

$$x = f(t) \qquad y = g(t) \qquad z = h(t)$$

as t varies over ℓ is called a **space curve**.

13.2 Derivatives and Integrals of Vector Functions

Derivatives

The derivative is defined analogously to limits

$$r'(t) = \langle f'(t), g'(t), h'(t) \rangle$$

Derivative rules (e.g. Product Rule, Chain Rule, etc.) hold similarly, where it suffices to apply each rule independently for each dimension. There are, however, a few differences.

Let $f(t)$ be a real-valued function, and u, v be vector-valued functions Then,

$$\frac{\mathrm{d}}{\mathrm{d}t}[f(t)u(t)] = f'(t)u(t) + f(t)u'(t)$$

And,

$$\frac{\mathrm{d}}{\mathrm{d}t}[u(t) \cdot v(t)] = u'(t) \cdot v(t) + u(t) \cdot v'(t)$$
$$\frac{\mathrm{d}}{\mathrm{d}t}[u(t) \times v(t)] = u'(t) \times v(t) + u(t) \times v'(t)$$

Note the difference between the two. (The second set of differential equations defines the derivative of the **dot product** and **cross product**, respectively).

Integrals

Again, the integral is defined analogously

$$\int_a^b r(t)\,dt = \left\langle \left(\int_a^b f(t)\,dt\right) \left(\int_a^b g(t)\,dt\right), \left(\int_a^b h(t)\,dt\right) \right\rangle$$

The Fundamental Theorem of Calculus extends to vector functions too.

$$\int_a^b r(t)\,dt = R(t)|_a^b = R(b) - R(a)$$

Where R is the antiderivative of r, i.e. $R'(t) = r(t)$.

13.3 Arc Length and Curvature

Length of a Curve

The formula for 2 dimensional arc length extends simply to $\mathbb{R}^3$.

$$L = \int_a^b \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}z}{\mathrm{d}t}\right)^2}\,dt$$

More succinctly and extending to $\mathbb{R}^n$,

Arc Length of a Vector-Valued Function

$$L = \int_a^b |r'(t)|\,dt$$

Note that, from this, we can also write an expression for the **arc length function** $s(t)$ as

$$s(t) = \int_a^t |r'(u)|\,du$$

Then,

$$\frac{\mathrm{d}s}{\mathrm{d}t} = |r'(t)|$$

Oftentimes, we can actually parametrize a curve in terms of **arc length** instead of time. Essentially, solve for $s(t)$ in terms of t, then solve for t in terms of $s(t)$, which gives the function $t(s)$. Substitute $r(t) \to r(t(s))$ to get a parameterization of r in terms of s.

Curvature

A curve parameterized by $r(t)$ is **smooth** on a n interval ℓ if r' is continuous and $r'(t) \neq 0$ across ℓ. The curvature κ is a measure of how quickly the curve changes direction at time t or arc length s (depending on your parameterization). Specifically, it is the magnitude of the rate of change of the unit tangent vector with respect to some parameter.

- A large κ means the change in direction is sudden/quicker.

- A small κ means the change in direction is gradual.
- $\kappa = 0$ means there is no curvature at all, i.e. a straight line.

First, the expression for the unit tangent vector $T(t)$ at time t:

$$T(t) = \frac{r'(t)}{|r'(t)|}$$

Therefore, we can write that the curvature of a curve is

$$\kappa = \frac{\mathrm{d}T}{\mathrm{d}s}$$

This expression would be easier to use with the parameter t, so use chain rule to write

$$\begin{aligned}\frac{\mathrm{d}T}{\mathrm{d}s} &= \frac{\frac{\mathrm{d}T}{\mathrm{d}t}}{\frac{\mathrm{d}s}{\mathrm{d}t}} \\ &= \frac{T'(t)}{r'(t)}\end{aligned}$$

Thus,

Curvature of a Vector-Valued Function

$$\kappa(t) = \frac{|T'(t)|}{|r'(t)|}$$

There is an additional formula we can derive, actually, that is often more convenient. Namely,

$$\kappa(t) = \frac{|r'(t) \times r''(t)|}{|r'(t)|^3}$$

Proof

Remember that $T(t) = \frac{r'(t)}{|r'(t)|}$ and $|r'(t)| = \frac{\mathrm{d}s}{\mathrm{d}t}$. Thus,

$$r'(t) = |r'(t)|T(t) = \frac{\mathrm{d}s}{\mathrm{d}t}T(t) \qquad (1)$$

Deriving,

$$r''(t) = \frac{\mathrm{d}^2 s}{\mathrm{d}t^2}T(t) + \frac{\mathrm{d}s}{\mathrm{d}t}T'(t) \qquad (2)$$

Taking the cross product of $(1) \times (2)$, and using the fact that $v \times v = 0$ for any vector v,

$$r'(t) \times r''(t) = \left(\frac{\mathrm{d}s}{\mathrm{d}t}\right)^2 (T(t) \times T'(t))$$

Taking the norm of both sides gives

$$|r'(t) \times r''(t)| = \left(\frac{\mathrm{d}s}{\mathrm{d}t}\right)^2 |T(t) \times T'(t)| \qquad (3)$$

$T(t)$ and $T'(t)$ are orthogonal (this fact is proven in the next section), so

$$
\begin{aligned}
T(t) \times T'(t) &= |T(t)||T''(t)| \sin\theta \\
&= |T(t)||T''(t)| \sin\frac{\pi}{2} \\
&= |T(t)||T''(t)|
\end{aligned}
$$

Substituting into (3),

$$
\begin{aligned}
|r'(t) \times r''(t)| &= \left(\frac{\mathrm{d}s}{\mathrm{d}t}\right)^2 |T(t)||T'(t)| \\
&= \left(\frac{\mathrm{d}s}{\mathrm{d}t}\right)^2 |T'(t)| && \text{since } |T(t)| = 1 \\
|T'(t)| &= \frac{|r'(t) \times r''(t)|}{\left(\frac{\mathrm{d}s}{\mathrm{d}t}\right)^2} \\
&= \frac{|r'(t) \times r''(t)|}{|r'(t)|^2} \\
\kappa &= \frac{|T'(t)|}{|r'(t)|} \\
&= \frac{|r'(t) \times r''(t)|}{|r'(t)|^3}
\end{aligned}
$$

The Normal and Binormal Vectors

At any arbitrary point on a smooth space curve $r(t)$, there exist many vectors orthogonal to the unit tangent vector $T(t)$. One particular vector, $T'(t)$, is actually always orthogonal to $T(t)$. The proof follows below for interested readers.

Proof of Orthogonality between $T(t)$ and $T'(t)$

Consider any vector-valued function $r(t)$ such that $|r(t)| = c, \forall t$ where c is some constant. Also, for orthogonality to be true, note that $r(t) \cdot r'(t) = 0, \forall t$.
Now, consider the following:

$$
r(t) \cdot r(t) = |r(t)|^2 = c^2, \forall t
$$

In other words, $\frac{\mathrm{d}}{\mathrm{d}t}[r(t) \cdot r(t)] = 0, \forall t$, since it is always c^2. We can write another expression for this derivative, though.

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}[r(t) \cdot r(t)] &= r'(t) \cdot r(t) + r(t) \cdot r'(t) \\
&= 2r'(t)r(t) && = 0
\end{aligned}
$$

Therefore, $r'(t) \cdot r(t) = 0, \forall t$, and $r'(t)$ is orthogonal to $r(t)$. Since $|T(t)| = 1, \forall t$, this proof holds for $T(t)$ and $T'(t)$.

Note, however, that $T'(t)$ is not necessarily a unit vector. Thus, at any point where $\kappa \neq 0$, we can define the **principal unit normal vector** $N(t)$ (also denoted **unit normal**). Note that $\kappa \neq 0$ is is a necessary condition, as $\kappa = 0$ implies $T'(t) = 0$, which would result in division by 0 below.

Principal Unit Normal Vector

$$
N(t) = \frac{T'(t)}{|T'(t)|}
$$

The rest of this section is not in scope for the class, but was just from the textbook and written for completeness.

Then, we also define a **binormal** vector $B(t)$ that is perpendicular to both $T(t)$ and $N(t)$:

Binormal Vector

$$B(t) = T(t) \times N(t)$$

The plane determined by N and B at a point P on curve C (sorry for all the notation D:) is called the **normal plane** of C at P, and consists of all lines orthogonal to the tangent vector T.

The plane determined by T and N at point P on curve C is called the **osculating plane** of C at P. It is the plane that most approximately contains the surrounding portions of the curve at P.

The circle that...

- lies in the osculating plane of C at P
- has the same tangent as C at P ($T(t)$)
- lies on the concave side of C (toward the direction in which N points)
- has radius $\rho = \frac{1}{\kappa}$ (giving it equivalent curvature, i.e. $\kappa_{\text{circle}} = \frac{1}{\rho} = \kappa$) *
  is called the **osculating circle** or **circle of curvature** of C at P. Essentially, this circle best approximates how C behaves near P, sharing the same tangent, normal, and curvature at P.

* *proof of the formula for the curvature of a circle is left as an exercise to the reader :)*

Reference

Formulas

$$L = \int_a^b |r'(t)|\, dt$$

$$T(t) = \frac{r'(t)}{|r'(t)|} \qquad N(t) = \frac{T'(t)}{|T'(t)|} \qquad B(t) = T(t) \times N(t)$$

$$\kappa = \left|\frac{\mathrm{d}T}{\mathrm{d}s}\right| = \frac{|T'(t)|}{|r'(t)|} = \frac{|r'(t) \times r''(t)|}{|r'(t)|^3}$$

Chapter 14: Partial Derivatives

14.1 Functions of Several Variables

First several pages of this chapter are just boring explanations of two-variable functions...

Level Curves

The level curves of a function f of two variables are the curves with equations $f(x, y) = k$, where k is a constant (in the range of f).

14.2 Limits and Continuity

Limits of Multivariate Functions

For a two-variable function,

$$\lim_{(x,y)\to(a,b)} f(x, y) = L$$

The definition extends similarly to higher dimensions.

Formally, $(\epsilon - \delta)$,

Limit Definition

Let f be a function of two variables whose domain D includes points arbitrarily close to (a, b). The **limit** of $f(x, y)$ as (x, y) approaches (a, b) is L if for every number $\epsilon > 0$ there is a corresponding number $\delta > 0$ such that if $(x, y) \in D$ and $0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta$ then $|f(x, y) - L| < \epsilon$.

In other words, the distance between $f(x, y)$ and L can be made infinitesimal by making the distance from (x, y) to (a, b) infinitesimal.

There is one complication with limits for multivariate functions that does not appear in univariate functions, though. Namely, (a, b) can be approached by (x, y) in any direction, infinitely many times. Thus, if there exist *more than one direction* that (a, b) is approached from and that causes $f(x, y)$ to approach *more than one distinct value*, the limit of $f(x, y)$ at (a, b) **does not exist**. Formally,

Limit Existence

If $f(x, y) \to L_1$ as $(x, y) \to (a, b)$ along a path C_1 and $f(x, y) \to L_2$ as $(x, y) \to (a, b)$ along a path C_2, where $L_1 \neq L_2$, then $\lim_{(x,y)\to(a,b)} f(x, y)$ does not exist.

Continuity

A two-variable function f is **continuous** at (a, b) if

$$\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b)$$

Also, if f is continuous at every point (a, b) in its domain D, then f is **continuous on** D.

Using the properties of limits, it is important to note the following property of continuous functions:

A function h produced by $f * g$, where $*$ is one of $+, -, \times, \div$, is continuous if and only if f, g are continuous.

As a result of this fact, all multivariate polynomials, as well as rational functions, are continuous.

Continuity for Proving Limits

If a function is continuous at (a, b), then the limit exists at (a, b). Then, for a function that is a composition of continuous functions, the limit exists at (a, b). This is often helpful for proving that a limit exists at (a, b) for a function! (So you don't have to use an $\epsilon - \delta$ proof).

14.3 Partial Derivatives

Partial Derivatives

A partial derivative of a multivariate function f is essentially just a univariate derivative that considers all other variables constant. Consider a two-variable function $f(x, y)$. Then, the partial derivative with respect to x at (a, b) is denoted $f_x(a, b)$ and

$$f_x(a, b) = g'(a)$$

Where $g(x) = f(x, b)$.

Similarly, the partial derivative with respect to y is

$$f_y(a, b) = h'(b)$$

Where $h(y) = f(a, y)$.

We can also write this in limit definition form.

$$f_x(a,b) = \lim_{h \to 0} \frac{f(a+h,b) - f(a,b)}{h}$$
$$f_y(a,b) = \lim_{h \to 0} \frac{f(a,b+h) - f(a,b)}{h}$$

So, TL;DR, when finding the partial derivative of a multivariate function f with respect to some variable, e.g. x, consider all other variables constant when deriving.

With regards to notation, the partial derivative of a multivariate function f with respect to x is

$$\frac{\partial f}{\partial x}$$

Higher-Order Derivatives

The following are all second partial derivatives of f:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)$$
$$f_{yx} = \frac{\partial^2 f}{\partial y \cdot \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$$
$$f_{xy} = \frac{\partial^2 f}{\partial x \cdot \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)$$
$$f_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)$$

Third, fourth, etc. partial derivatives of f are defined similarly.

Most functions we will see will actually have $f_{yx} = f_{xy}$. The following theorem specifies when this occurs:

> Clairaut's Theorem
>
> *Suppose f is defined on a disk D that contains the point (a, b). If the functions f_{xy} and f_{yx} are both continuous on D, then $f_{xy}(a, b) = f_{yz}(a, b)$.*

14.4 Tangent Planes and Linear Approximations

Tangent Planes

Suppose a surface S has equation $z = f(x, y)$, where f has continuous first partial derivatives, and let $P(x_0, y_0, z_0)$ be a point on S. Let C_1, C_2 be the curves that result from intersecting the surface S with the planes $y = y_0$ and $x = x_0$. Note that P lies on both C_1 and C_2. Le T_1 and T_2 be the tangent lines to curves C_1 and C_2 at P. Then the **tangent plane** to the surface S at the point P is the plane that contains both tangent lines T_1 and T_2.

Note that, for any other curve C that lies on the surface S, it passes through $P \iff$ its tangent line at P lies in the tangent plane. In other words, the tangent plane at P consists of all possible tangent lines at P, i.e. the tangent plane at P best approximates the surface S near P. This will be covered in more detail in 14.6.

Tangent Plane Derivation

A plane passing through $P(x_0, y_0, z_0)$ has the form

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

Let $a = -\frac{A}{C}$ and $b = -\frac{B}{C}$. Then,

$$z - z_0 = a(x - x_0) + b(y - y_0)$$

Consider the plane $y = y_0$, which intersects with S to form C_1. Substituting into this equation, we get

$$z - z_0 = a(x - x_0)$$

Remember that T_1 is the tangent line to the curve C_1, and lies in the tangent plane. Therefore, this above equation represents T_1. We know that the slope of tangent T_1 can be calculated as $f_x(x_0, y_0) = \frac{\partial f}{\partial x}$. Thus, $a = f_x(x_0, y_0)$. A similar argument follows for the plane $x = x_0$. Hence:

> **Tangent Plane Equation**
>
> Suppose f has continuous partial derivatives. The equation of the tangent plane to the surface $z = f(x, y)$ at $P(x_0, y_0, z_0)$ is
>
> $$z = z_0 + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Linear Approximations

Note that the tangent plane equation essentially represents an approximation of the function f at a point P that is a *linear function*. This function L is called the *linearization* of f at P, and the approximation $f \approx L$ is the linear approximation/tangent plane approximation of f at P.

Differentiability

Formally,

> **Differentiability**
>
> If $z = f(x, y)$, then f is differentiable at (a, b) if Δz can be expressed in the form
>
> $$\Delta z = f_x(a, b)\Delta x + f_y(a, b)\Delta y + \epsilon_1 \Delta x + \epsilon_2 \Delta y$$
>
> where $\epsilon_1, \epsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0, 0)$

In words,

> Differentiability
>
> *If the partial derivatives f_x and f_y exist near (a, b) and are continuous at (a, b), then f is differentiable at (a, b).*

We may also write the following

> **Differentiability (Limit Definition)**
>
> f is differentiable at (a, b) if
>
> $$\lim_{(x,y)\to(a,b)} \frac{f(x, y) - h(x, y)}{|(x, y) - (a, b)|} = 0$$

where $h(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$, i.e. is the linear approximation of f.

Differentials

For a univariate function $y = f(x)$, we can write

$$f'(x) = \frac{\mathrm{d}y}{\mathrm{d}x} \implies \mathrm{d}y = f'(x)\mathrm{d}x$$

For a two-variable function $z = f(x,y)$, we can instead write

$$\mathrm{d}z = f_x(x,y)\mathrm{d}x + f_y(x,y)\mathrm{d}y = \frac{\partial z}{\partial x}\mathrm{d}x + \frac{\partial z}{\partial y}\mathrm{d}y$$

For a general multivariate function $f(x_1, x_2, \ldots, x_n)$,

$$\mathrm{d}f = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\mathrm{d}x_i$$

14.5 Chain Rule

Chain Rule

Recall the chain rule applied to univariate functions $y = f(x)$ and $x = g(t)$:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\mathrm{d}y}{\mathrm{d}x} \cdot \frac{\mathrm{d}x}{\mathrm{d}t}$$

For multivariate functions, there actually exist several versions of the chain rule. First, we consider the case where x and y are univariate functions.

The Chain Rule (1)

Suppose that $z = f(x,y)$ is a differentiable function of x and y, where $x = g(t)$ and $y = h(t)$ are both differentiable functions in terms of t. Then z is a differentiable function of t and

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}$$

Now, we consider the case where x and y are multivariate functions.

The Chain Rule (2)

Suppose that $z = f(x,y)$ is a differentiable function of x and y, where $x = g(s,t)$ and $y = h(s,t)$ are differentiable functions of s and t. Then

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s} \qquad \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

In other words, just apply the case 1 chain rule separately to s and t.

The Chain Rule (General)

Suppose u is a differentiable function of the n variables $x_1, x_2, \dots, x_n$ and each x_i is a differentiable function of the m variables $t_1, t_2, \dots, t_m$. Then u is af function of $t_1, t_2, \dots, t_m$ and

$$\frac{\partial u}{\partial t_i} = \sum_{j=1}^{n} \frac{\partial u}{\partial x_j} \frac{\partial x_j}{\partial t_i}$$

Implicit Differentiation

Suppose that an equation of the form $F(x, y) = 0$ defines y implicitly as a differentiable function of x, i.e. $y = f(x)$ where $F(x, f(x)) = 0, \forall x \in \mathrm{Domain}(f)$. If F is differentiable, then we can apply Chain Rule to differentiate $F(x, y) = 0$ with respect to x:

$$\begin{aligned} \frac{\partial F}{\partial x} \cancelto{1}{\frac{\mathrm{d}x}{\mathrm{d}x}} + \frac{\partial F}{\partial y} \frac{\mathrm{d}y}{\mathrm{d}x} &= 0 \\ \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} \frac{\mathrm{d}y}{\mathrm{d}x} &= 0 \\ \frac{\partial F}{\partial y} \frac{\mathrm{d}y}{\mathrm{d}x} &= -\frac{\partial F}{\partial x} \\ \frac{\mathrm{d}y}{\mathrm{d}x} &= \boxed{-\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}}} \end{aligned}$$

Now suppose that z is given implicitly as a function $z = f(x, y)$ by an equation of the form $F(x, y, z) = 0$, i.e. $F(x, y, f(x, y)) = 0, \forall (x, y) \in \mathrm{Domain}(f)$. If F and f are differentiable, then we can apply Chain Rule to differentiate the equation $F(x, y, z) = 0$ with respect to x:

$$\begin{aligned} \frac{\partial F}{\partial x} \cancelto{1}{\frac{\partial x}{\partial x}} + \frac{\partial F}{\partial y} \cancelto{0}{\frac{\partial y}{\partial x}} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial x} &= 0 \\ \frac{\partial F}{\partial z} \frac{\partial z}{\partial x} &= -\frac{\partial F}{\partial x} \\ \frac{\partial z}{\partial x} &= \boxed{-\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}}} \end{aligned}$$

Similarly, by differentiating with respect to y, we derive

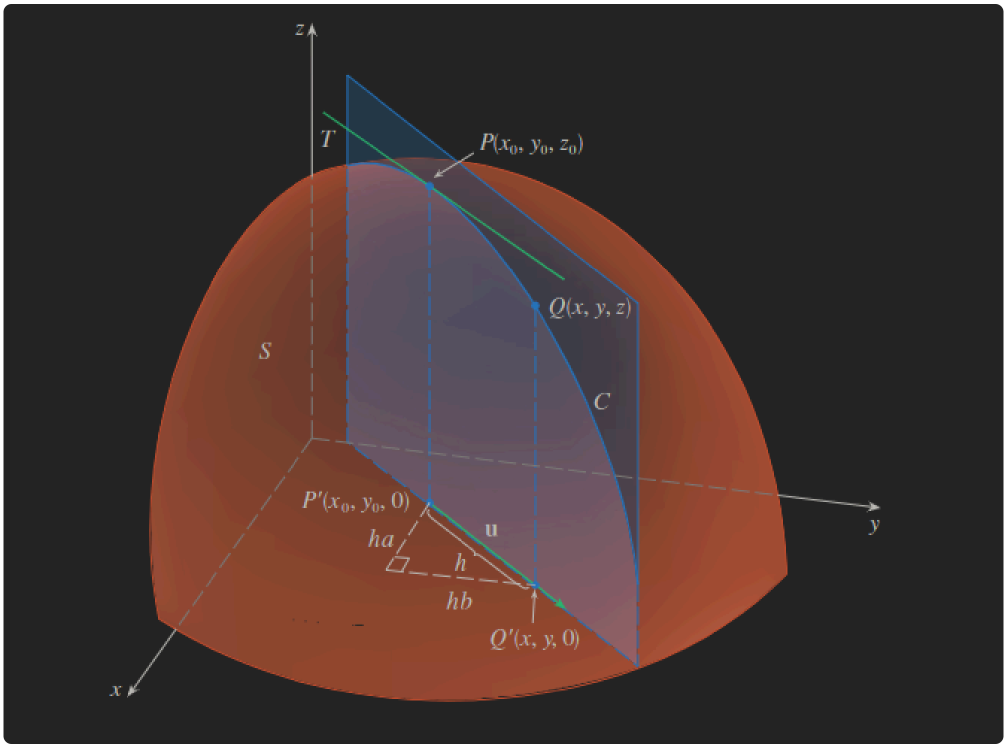
$$\frac{\partial z}{\partial y} = \boxed{-\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}}}$$

⚠ As a sidenote, the textbook mentions that the Implicit Function Theorem as stipulating conditions under which these are valid. These are not included in the notes because they seemed to be rather irrelevant to the necessary knowledge for Math 53.

14.6 Directional Derivatives and the Gradient Vector

Directional Derivative

The following diagram may be useful to reference as you read this section:



Consider $z = f(x, y)$. We aim to find the rate of change of z at (x_0, y_0) in the direction of the arbitrary unit vector $\mathbf{u} = \langle a, b \rangle$.

Consider the surface described by z, and let $z_0 = f(x_0, y_0)$. Then $P(x_0, y_0, z_0)$ lies on S. The vertical plane that passes through P in the direction of $\mathbf{u}$ intersects S in a curve C. The slope of the tangent line T to C at the point P is the rate of change of z in the direction of $\mathbf{u}$.

Let $Q(x, y, z)$ be another point on C. Let P', Q' be the projections of P, Q onto the xy-plane. Then, the vector $\overrightarrow{P'Q'} \parallel \mathbf{u}$, and thus

$$\overrightarrow{P'Q'} = h\mathbf{u} = \langle ha, hb \rangle$$

for some scalar h. In other words, $x - x_0 = ha$ and $y - y_0 = hb$, i.e. $x = x_0 + ha$ and $y = y_0 + hb$. Therefore,

$$\frac{\Delta z}{h} = \frac{z - z_0}{h} = \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}$$

Taking the limit as $h \to 0$, we get

ⓘ Directional Derivative (Limit Definition)

The directional derivative of f at (x_0, y_0) in the direction of the unit vector $\mathbf{u} = \langle a, b \rangle$ is

$$D_{\mathbf{u}} f(x_0, y_0) = \lim_{h \to 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}$$

(*if the limit exists*)

We can rewrite this in a more useful form:

ⓘ Directional Derivative (Derivative Definition)

If f is a differentiable function of x, y, then f has a directional derivative in the direction of any unit vector $\mathbf{u} = \langle a. b \rangle$ and

$$D_{\mathbf{u}} f(x, y) = f_x(x, y)a + f_y(x, y)b$$

The above follows directly from Chain Rule, and its proof is left as an exercise to the reader. (Hint: consider deriving the function $g(h) = f(x_0 + ha, y_0 + hb)$).

Also, if $\mathbf{u}$ make an angle θ with the positive x-axis, we can write $\mathbf{u} = \langle \cos\theta, \sin\theta \rangle$. Then, the formula becomes

$$D_u f(x, y) = f_x(x, y)\cos\theta + f_y(x, y)\sin\theta$$

Gradient Vector

Note that the directional derivative of $f(x, y)$ can actually be written as a dot product:

$$\begin{aligned} D_u f(x, y) &= f_x(x, y)a + f_y(x, y)b \\ &= \langle f_x(x, y), f_y(x, y) \rangle \cdot \langle a, b \rangle \\ &= \langle f_x(x, y), f_y(x, y) \rangle \cdot \mathbf{u} \end{aligned}$$

The first vector in the dot product appears frequently in many contexts, and so is specially denoted as follows:

Gradient

If f is a function of two variables x, y, then the **gradient** of f is the vector function ∇f defined by

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j}$$

We may then rewrite the directional derivative equation again.

Directional Derivative (Gradient Definition)

$$D_u f(x, y) = \nabla f(x, y) \cdot \mathbf{u}$$

In other words, the directional derivative in the direction of $\mathbf{u}$ is the scalar projection of the gradient vector onto $\mathbf{u}$. (Recall $\text{comp}_a b = \frac{a \cdot b}{|a|}$, and that $\mathbf{u}$ is a unit vector).

Maximizing the Directional Derivative

We aim to find the maximal directional derivative of f at a given point, i.e. the direction in which f changes the fastest at a given point and the corresponding magnitude of rate of change.

Note that

$$\begin{aligned} D_u f &= \nabla f \cdot \mathbf{u} \\ &= |\nabla f||u|\cos\theta \\ &= |\nabla f|\cos\theta \end{aligned}$$

where θ denotes the angle between ∇f and $\mathbf{u}$. The maximum value of $\cos\theta$ is 1, and occurs when $\theta = 0$. Therefore, the maximum value of $D_u f$ is $|\nabla f|$, and it occurs when $\theta = 0 \implies \mathbf{u} \parallel \nabla f$, i.e. the two vectors have the same direction. Thus,

Maximal Directional Derivative

The maximum value of the directional derivative $D_u f$ is $|\nabla f|$ and occurs when $\mathbf{u} \parallel \nabla f$.

Tangent Planes to a Level Surface

Suppose S is a surface with equation $F(x, y, z) = k$, i.e. it is a level surface of a three-variable function F. Let $P(x_0, y_0, z_0)$ lie on S and let C be an arbitrary curve that lies on S and passes through P. Let $C = \mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$, and let $t_0 \in \mathbb{R}$ such that $P = \mathbf{r}(t_0)$. Consider that

$$\begin{aligned} F(x(t), y(t), z(t)) &= k \\ \frac{\partial}{\partial t} F(x(t), y(t), z(t)) &= \frac{\partial}{\partial t} k \\ \frac{\partial F}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial F}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{\partial F}{\partial z}\frac{\mathrm{d}z}{\mathrm{d}t} &= 0 \\ \langle F_x, F_y, F_z\rangle \cdot \langle x'(t), y'(t), z'(t)\rangle &= 0 \\ \nabla F \cdot \mathbf{r}'(t) &= 0 \end{aligned}$$

Now, substituting $t = t_0$,

$$\nabla F(x_0, y_0, z_0) \cdot \mathbf{r}'(t_0) = 0$$

In other words,

Theorem

The gradient vector at P of $F(x, y, z)$ is ***perpendicular*** *to the tangent vector $\mathbf{r}'(t_0)$ to any curve C on S that passes through P.*

If $\nabla F(x_0, y_0, z_0) \neq \mathbf{0}$, we can then define the **tangent plane to the level surface** at $P(x_0, y_0, z_0)$ as the plane that passes through P and has normal vector $\nabla F(x_0, y_0, z_0)$. Thus,

Tangent Plane to a Level Surface

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0$$

We can also define a line called the **normal line** to S at P such that it passes through P and is perpendicular to the tangent plane. The direction of this normal line is clearly the same as the gradient vector, and as such is defined as

Normal Line

$$\frac{x - x_0}{F_x(x_0, y_0, z_0)} = \frac{y - y_0}{F_y(x_0, y_0, z_0)} = \frac{z - z_0}{F_z(x_0, y_0, z_0)}$$

14.7 Maximum and Minimum Values

A two-variable function f has a **local maximum** at (a, b) if $f(x, y) \leq f(a, b)$ when (x, y) is near (a, b). A **local minimum** is defined analogously.

If $f(x, y) \leq f(a, b), \forall (x, y) \in D$, the domain of $f(x, y)$,then $f(a, b)$ is an **absolute maximum**. An **absolute minimum** is defined analogously.

There exist the following tests for finding the local extrema of $f(x, y)$, which are analogous to their univariate counterparts.

First Derivative Test

If f has a local extrema at (a, b) and the first order partial derivatives exist there, then $f_x(a, b) = f_y(a, b) = 0$. This is considered a **critical point**.

Second Derivative Test

Suppose the second partial derivatives of f are continuous on a disk with center (a, b), and (a, b) is a critical point of f. Let

$$D(a, b) = f_{xx}(a, b) f_{yy}(a, b) - [f_{xy}(a, b)]^2 = \begin{matrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{matrix}$$

Then,
(1) $D > 0$ and $f_{xx}(a, b) > 0 \implies f(a, b)$ is a local minimum.
(2) $D > 0$ and $f_{xx}(a, b) < 0 \implies f(a, b)$ is a local maximum.
(3) $D < 0 \implies f(a, b)$ is not a local extrema. Instead, (a, b) is a saddle point of f.
(4) $D = 0 \implies$ no information about $f(a, b)$.

Meanwhile, we also consider the absolute extrema of $f(x, y)$ over some closed interval. For $\mathbb{R}^2$, a **bounded set** is one that is contained in some disk. Meanwhile, a **closed set** is one that additionally contains all its boundary points. With these definitions, we then define the Extreme Value Theorem for two-variable functions.

Extreme Value Theorem

If f is continuous on a closed, bounded set D in $\mathbb{R}^2$, then f attains an absolute maximum value $f(x_1, y_1)$ and an absolute minimum value $f(x_2, y_2)$ at some points (x_1, y_1) and (x_2, y_2) in D.

To find the absolute extrema, we perform the following process.

Finding Absolute Extrema

1. Find the values of f at the critical points in D.
2. Find the extreme values of f on the boundary of D.

Notice the similarity to the univariate analogue! However, one major difference is that, when determining the extreme values on the boundary of D, it is not possible to calculate the values of every point on the boundary of D (since there are infinitely many!). Instead, it suffices to set constraints on x, y such that (x, y) is on the boundary, and determine the extreme values of $f(x, y)$ over these constraints, typically through taking the derivatives of the consequent univariate functions. It may be necessary to split the boundary up into multiple different constraints. See the textbook for some helpful examples!

14.8 Lagrange Multipliers

One Constraint

Lagrange's method is a way of maximizing/minimizing a general function $f(x, y, z)$ when a constraint of the form $g(x, y, z) = k$ is considered.

Consider the function $f(x, y, z)$ subject to the constraint $g(x, y, z) = k$. In other words, (x, y, z) is restricted to lie on the level surface S with equation $g(x, y, z) = k$.

Suppose f has an extreme value at a point $P(x_0, y_0, z_0)$ on the surface S and let C be a curve with vector equation $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ that lies on S and passes through P. If t_0 is the parameter value corresponding to the point P, then $\mathbf{r}(t_0) = \langle x_0, y_0, z_0 \rangle$. The composite function $h(t) = f(x(t), y(t), z(t))$ represents the values that f takes on the curve C. Since f has an extreme value at (x_0, y_0, z_0), which corresponds to t_0 for $\mathbf{r}(t)$, h has an extreme value at t_0, i.e. $h'(t_0) = 0$. Since f is differentiable, we can apply Chain Rule as follows:

$$\begin{aligned} 0 &= h'(t_0) \\ &= f_x(x_0, y_0, z_0)x'(t_0) + f_y(x_0, y_0, z_0)y'(t_0) + f_z(x_0, y_0, z_0)z'(t_0) \\ &= \nabla f(x_0, y_0, z_0) \cdot \mathbf{r}'(t_0) \end{aligned}$$

In other words, the gradient vector $\nabla f(x_0, y_0, z_0)$ is orthogonal to the tangent vector $\mathbf{r}'(t_0)$ to every such curve C. From section 14.6, we know that the gradient vector of g, $\nabla g(x_0, y_0, z_0)$, is also orthogonal to $\mathbf{r}'(t_0)$ for every curve C, since $g(x, y, z) = k$ describes the level surface S. In other words, $\nabla f(x_0, y_0, z_0) \parallel \nabla g(x_0, y_0, z_0)$. Further, if $\nabla g(x_0, y_0, z_0) \neq 0$, then $\exists \lambda$ such that

$$\nabla f(x_0, y_0, z_0) = \lambda \, \nabla g(x_0, y_0, z_0)$$

λ is called the **Lagrange multiplier**.

> **Method of Lagrange Multipliers**
>
> To find the extreme values of $f(x, y, z)$ subject to the constraint $g(x, y, z) = k$ (assuming these extreme values exist and $\nabla g \neq 0$ on the surface $g(x, y, z) = k$).
>
> 1. Find all values of (x, y, z) and λ such that $\nabla f(x, y, z) = \lambda \, \nabla g(x, y, z)$ and $g(x, y, z) = k$.
> 2. Evaluate f at all points (x, y, z) from step 1. The largest of these values is the maximum value of f, and analogously for the minimum value of f.

For a 3-variable function, we can write the vector equation $\nabla f = \lambda \, \nabla g$ in terms of its components to transform the equation in step 1 into

$$f_x = \lambda g_x \qquad f_y = \lambda g_y \qquad f_z = \lambda g_z \qquad g(x, y, z) = k$$

This is a system of four unknowns and four equations, so it is solvable!

Two Constraints

We can also find the extreme values of $f(x, y, z)$ subject to two constraints $g(x, y, z) = k$ and $h(x, y, z) = c$. Then, we can write

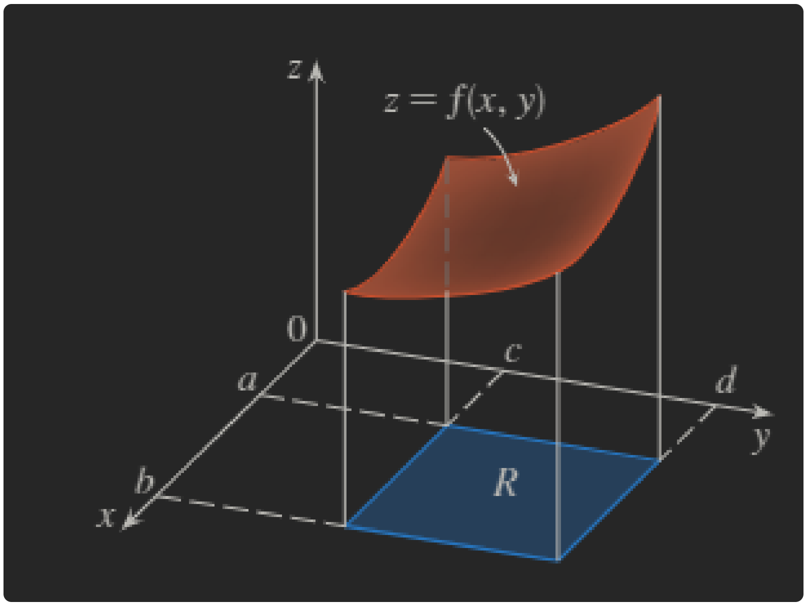
$$\nabla f(x_0, y_0, z_0) = \lambda \, \nabla g(x_0, y_0, z_0) + \mu \, \nabla h(x_0, y_0, z_0)$$

Chapter 15: Multiple Integrals

15.1 Double Integrals over Rectangles

Derivation

Consider a function f of two variables bounded by a closed rectangle R, i.e. bounded by $a \leq x \leq b$ and $c \leq y \leq d$, for some $a, b, c, d \in \mathbb{R}$. Then, define the 3D solid S such that it lies between the rectangle R lying in the xy-plane and the surface with equation $z = f(x, y)$, bounded by the above bounds. The below image may help you understand this:



We aim to find the volume of S.

Consider dividing rectangle R into subrectangles, analogous to how we might do a Riemann sum, but instead in 3 dimensions. For any subrectangle, we can approximate the volume of S lying above the subrectangle by computing $f(x_0, y_0)\Delta A$, where x_0, y_0 represents an arbitrary point in the subrectangle and ΔA represents the area of the subrectangle.

Approximating the volume mathematically, we get

$$V \approx \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_i, y_j)\Delta A_{i,j}$$

Where m, n denote the number of subdivisions in each dimension (x-axis and y-axis).

To find the true volume, we of course just take the limit:

$$V = \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_i, y_j)\Delta A_{i,j}$$

Double Integral

Mathematically, we often write instead that

> **Double Integral**
>
> $$V = \iint_R f(x,y)\,dA = \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_i, y_j)\Delta A_{i,j}$$

We don't have to use the limit definition to integrate, however. Instead, we can express a double integral, as an iterated integral:

$$\int_c^d \int_a^b f(x,y)\,dx\,dy = \int_c^d \left[\int_a^b f(x,y)\,dx\right] dy$$
$$\int_a^b \int_c^d f(x,y)\;\mathrm{d}y\,dx = \int_a^b \left[\int_c^d f(x,y)\,dy\right] dx$$

Essentially, we can integrate independently of the other integrals. Note also that **partial integration** works analogously as to partial differentiation, i.e. you treat other variables as constant.

Additionally, note that many functions f can be integrated in either order when finding the volume of the solid S.

Fubini's Theorem

If f is continuous over the rectangle $R = \{(x, y) \mid a \le x \le b, c \le y \le d\}$

$$\iint\limits_R f(x, y)\ \mathrm{d}A = \int_c^d \int_a^b f(x, y)\,\mathrm{d}x\,\mathrm{d}y = \int_a^b \int_c^d f(x, y)\,\mathrm{d}y\,\mathrm{d}x$$

Exception: Product of Univariate Functions

In the special case where $f(x, y) = g(x)h(y)$, i.e. can be factored as a product of a function of only x and a function of only y, then we can write

$$\iint\limits_R f(x, y)\,\mathrm{d}A = \iint\limits_R g(x)h(y)\,\mathrm{d}A = \int_a^b g(x)\,\mathrm{d}x \int_c^d h(y)\,\mathrm{d}y$$

We can do this because y is constant when integrating with respect to x, and vice versa.

Average Value

This is calculated analogously to how it is in 1 dimension. Simply,

$$f_{\text{avg}} = \frac{1}{A(R)} \iint\limits_R f(x, y)\,\mathrm{d}A$$

Where $A(R)$ is the area of the rectangle R. This should make sense intuitively.

15.2 Double Integrals over General Regions

We aim to integrate volumes of solids over arbitrary plane regions, not just rectangles. We will consider two types of regions, and their corresponding integrations.

In order to evaluate $\iint_D f(x, y)\,\mathrm{d}A$, we consider a rectangle $R = [a, b] \times [c, d]$ that contains the plane region D, i.e. $D \subseteq R$. We define a function F with domain R as

$$F(x, y) = \begin{cases} f(x, y) & \text{if } (x, y) \text{ is in } D \\ 0 & \text{if } (x, y) \text{ is in } R \text{ but not in } D \end{cases}$$

Then, if F is integrable over R, we have

$$\iint\limits_D f(x, y)\,\mathrm{d}A = \iint\limits_R F(x, y)\,\mathrm{d}A$$

Type I

A plane region D is said to be of **type I** if it lies between the graphs of two continuous functions of x, that is:

$$D = \{(x, y) \mid a \le x \le b, g_1(x) \le y \le g_2(x)\}$$

Then, we may write

$$\iint\limits_D f(x, y)\,\mathrm{d}A = \iint\limits_R F(x, y)\,\mathrm{d}A = \int_a^b \int_c^d F(x, y)\,\mathrm{d}y\,\mathrm{d}x$$

Note that $F(x, y) = 0$ if $y < g_1(x)$ or $y > g_2(x)$. Thus,

$$\int_c^d F(x,y)\,\mathrm{d}y = \int_{g_1(x)}^{g_2(x)} F(x,y)\,\mathrm{d}y = \int_{g_1(x)}^{g_2(x)} f(x,y)\,\mathrm{d}y$$

Therefore,

> **Type I Volume**
>
> If f is continuous on a type I region D such that $D = \{(x,y) \mid a \le x \le b, g_1(x) \le y \le g_2(x)\}$, then
>
> $$\iint_D f(x,y)\,\mathrm{d}A = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,\mathrm{d}y\,\mathrm{d}x$$

Type II

A plane region D is said to be **type II** if it is between the graphs of two continuous functions of y, i.e. type I but x is bounded instead of y:

$$D = \{(x,y) \mid c \le y \le y, h_1(y) \le x \le h_2(y)\}$$

Thus, by similar derivation as in Type I, we find

> **Type II Volume**
>
> $$\iint_D f(x,y)\,\mathrm{d}A = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\,\mathrm{d}x\,\mathrm{d}y$$

Properties of Double Integrals

$$\begin{aligned}
&\textbf{(1)}\ \iint_D [f(x,y)+g(x,y)]\,\mathrm{d}A = \iint_D f(x,y)\,\mathrm{d}A + \iint_D g(x,y)\,\mathrm{d}A \\
&\textbf{(2)}\ \iint_D cf(x,y)\,\mathrm{d}A = c\iint_D f(x,y)\,\mathrm{d}A,\ \text{for some constant } c \\
&\textbf{(3)}\ f(x,y) \ge g(x,y), \forall (x,y) \in D \implies \iint_D f(x,y)\,\mathrm{d}A \ge \iint_D g(x,y)\,\mathrm{d}A \\
&\textbf{(4)}\ D = D_1 \cup D_2, D_1 \cap D_2 = \emptyset \implies \iint_D f(x,y)\,\mathrm{d}A = \iint_{D_1} f(x,y)\,\mathrm{d}A + \iint_{D_2} f(x,y)\,\mathrm{d}A \\
&\textbf{(5)}\ \iint_D 1\,\mathrm{d}A = A(D) \\
&\textbf{(6)}\ m \le f(x,y) \le M, \forall (x,y) \in D \implies mA(D) \le \iint_D f(x,y)\,\mathrm{d}A \le MA(D)
\end{aligned}$$

Where,

- **(4)** $D_1 \cap D_2$ can contain boundary points, but only that.
- **(5)** $A(D)$ is the area of the region D.

15.3 Double Integrals in Polar Coordinates

The proof of the formula for this will be left as an exercise to the reader since it is similar to the proof of double integrals over rectangles. Hint: consider dividing a polar region into polar subrectangles.

> **Double Integral over Polar Rectangle**
>
> If f is continuous on a polar rectangle R describe by $0 \le a \le r \le b, \alpha \le 0 \le \beta$, where $0 \le \beta - \alpha \le 2\pi$, then

$$\iint_R f(x,y)\,\mathrm{d}A = \int_\alpha^\beta \int_a^b f(r\cos\theta, r\sin\theta)\cdot r\,\mathrm{d}r\,\mathrm{d}\theta$$

⚠ Note the additional factor of r on the RHS. Careful not to forget this!

Generalizing to any polar region, similarly to how it was done in 15.2, we get

ⓘ Double Integral over General Polar Region

If f is continuous on a polar region of the form

$$D = \{(r,\theta) \mid \alpha \le \theta \le \beta\}, h_1(\theta) \le r \le h_2(\theta)$$

Then

$$\iint_D f(x,y)\,\mathrm{d}A = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta, r\sin\theta)\cdot r\,\mathrm{d}r\,\mathrm{d}\theta$$

15.4 Applications of Double Integrals

Density and Mass

Let $\rho(x,y)$ denote the **density function** of a region D. Thus,

$$\rho(x,y) = \lim \frac{\Delta m}{\Delta A}$$

Where $\Delta m, \Delta A$ are the mass and area of an infinitesimal rectangle containing (x,y). Thus, it naturally follows that

$$m = \iint_D \rho(x,y)\,\mathrm{d}A$$

Physicists consider other types of density that aren't mass as well. For instance, if the charge density if given by $\sigma(x,y)$ at (x,y) in D, then the total charge Q is simply

$$Q = \iint_D \sigma(x,y)\,\mathrm{d}A$$

Moments and Centers of Mass

The **moment** of a particle is the product of its mass and its directed distance from the axis. Consider an infinitesimal rectangle R_{xy} containing the point (x,y). Then, the mass of $R_{xy} \approx \rho(x,y)\Delta A$, and the moment of R_{xy} with respect to the x-axis is therefore

$$M_x(R_{xy}) = \rho(x,y)\Delta A \cdot y_{xy}$$

Taking the limit of the sum of these quantities gives us the moment of the entire *lamina* about the x-axis:

$$M_x = \iint_D y\rho(x,y)\,\mathrm{d}A$$

And about the y-axis,

$$M_y = \iint_D x\rho(x,y)\,\mathrm{d}A$$

Using these, we derive the coordinates $(\overline{x}, \overline{y})$ of the center of mass of a lamina occupying the region D:

$$\overline{x} = \frac{M_y}{m} = \frac{1}{m} \iint_D x\rho(x, y)\,\mathrm{d}A \qquad \overline{y} = \frac{M_x}{m} = \frac{1}{m} \iint_D y\rho(x, y)\,\mathrm{d}A$$

Where the mass m is given by

$$m = \iint_D \rho(x, y)\,\mathrm{d}A$$

Moment of Inertia

The **moment of inertia**, or **second moment**, of a particle of mass m about an axis is defined to be mr^2, where r is the distance from the particle to the axis. For a lamina with density function $\rho(x, y)$ and occupying a region D, we can similarly derive equations for this.

$$I_x = \iint_D y^2 \rho(x, y)\,\mathrm{d}A$$
$$I_y = \iint_D x^2 \rho(x, y)\,\mathrm{d}A$$

The moment of inertia **about the origin**, or the **polar moment of inertia**, is meanwhile

$$I_0 = \iint_D (x^2 + y^2)\rho(x, y)\,\mathrm{d}A = I_x + I_y$$

Moreover, we may define the **radius of gyration of a lamina about an axis** is the number R such that

$$mR^2 = I$$

In particular, we may define

$$m\overline{\overline{y}}^2 = I_x \qquad m\overline{\overline{x}}^2 = I_y$$

Probability

The **probability density function (PDF)** f of a continuous random variable X has the property that $f(x) \geq 0, \forall x$, and $\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1$. Additionally,

$$P(a \leq X \leq b) = \int_a^b f(x)\,\mathrm{d}x$$

Now, consider a pair of continuous random variables X, Y. The **joint density function** of X, Y is a function f of two variables such that the probability that (X, Y) lies in a region D is

$$P((X, Y) \in D) = \iint_D f(x, y)\,\mathrm{d}A$$

Note also that

$$\iint_{\mathbb{R}^2} f(x, y)\,\mathrm{d}A = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\,\mathrm{d}x\,\mathrm{d}y = 1$$

Expected Values

If X is a random variable with PDF f, then its mean is

$$\mu = \int_{-\infty}^{\infty} x f(x)\,\mathrm{d}x$$

If X, Y are random variables with joint density function f, we define the X**-mean** and Y**-mean**, i.e. the **expected values** of X, Y, to be

$$\mu_1 = \iint_{\mathbb{R}^2} xf(x,y)\,\mathrm{d}A \qquad \mu_2 = \iint_{\mathbb{R}^2} yf(x,y)\,\mathrm{d}A$$

15.5 Surface Area

Consider a point $P_{xy} = (x_i, y_j, f(x_i, y_j))$ on some surface S described by $z = f(x,y)$. Now, consider some infinitesimal rectangle R_{ij} lying in the xy-plane that contains the projection of P_{ij} down into the xy-plane. Then, we may approximate the surface area of the curve lying above this rectangle R_{ij} with the area of the parallelogram of the tangent plane T_{ij} at point P_{ij} bounded by the bounds of R_{ij} (in x and y). We will let one vertex of this parallelogram lie at point P_{ij} (i.e. P_{ij} lies directly above a vertex of R_{ij}). Let this area be denoted by ΔT_{ij}. Then, we may approximate the **surface area** of S as

$$A(S) = \lim_{m,n\to\infty} \sum_{i=1}^{m}\sum_{j=1}^{n} \Delta T_{ij}$$

Now, we aim to find an expression for ΔT_{ij}. Since it is the area of some parallelogram in T_{ij}, we can let $\mathbf{a}, \mathbf{b}$ be vectors that lie along the sides of this parallelogram. Additionally, let Δx and Δy be the (infinitesimal) side lengths of the rectangle R_{ij}. Then, we may define

$$\begin{aligned}\mathbf{a} &= \Delta x\mathbf{i} + f_x(x_i, y_j)\Delta x\mathbf{k}\\ \mathbf{b} &= \Delta y\mathbf{j} + f_y(x_i, y_j)\Delta y\mathbf{k}\end{aligned}$$

Note that we are enabled to do this because we can just let the sides of the rectangle R_{ij} be parallel to the x and y axes, and therefore one side of the parallelogram will have no component in the x direction, and similarly the other side will have no component in the y direction.

Since the area of the parallelogram with respect to $\mathbf{a}, \mathbf{b}$ is $|\mathbf{a}\times\mathbf{b}|$, we may write that

$$\Delta T_{ij} = |\mathbf{a}\times\mathbf{b}| = \det\begin{bmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ \Delta x & 0 & f_x(x_i,y_j)\Delta x\\ 0 & \Delta y & f_y(x_i,y_j)\Delta y\end{bmatrix}$$

Simplifying, we eventually get that

$$\Delta T_{ij} = \sqrt{[f_x(x_i,y_j)]^2 + [f_y(x_i,y_j)]^2 + 1}\,\Delta A$$

Where $\Delta A = \Delta x\Delta y$, i.e. the area of R_{ij}. And thus,

> **Surface Area**
>
> The area of the surface S with equation $z = f(x,y), (x,y)\in D$, where f_x, f_y are continuous, is
>
> $$\begin{aligned}A(S) &= \iint_D \sqrt{[f_x(x_i,y_j)]^2 + [f_y(x_i,y_j)]^2 + 1}\,\mathrm{d}A\\ &= \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\,\mathrm{d}A\end{aligned}$$

Note that the similarity between the surface area formula and the arc length formula may help with remembering this:

$$L = \int_a^b \sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x$$

15.6 Triple Integrals

First, let's consider the triple integral of a function f defined on a rectangular box

$$B = \{(x, y, z) \mid a \leq x \leq b, c \leq x \leq d, e \leq z \leq f\}$$

Similar to previous sections, we consider dividing B into infinitesimal sub-boxes and then finding the value of the triple integral by summing over the boxes' volumes.

$$\iiint\limits_{B} f(x, y, z)\,\mathrm{d}V = \lim_{l,m,n \to \infty} \sum_{i=1}^{l} \sum_{j=1}^{m} \sum_{k=1}^{n} f(x_{ijk}, y_{ijk}, z_{ijk}) \Delta V$$

Just like double integrals, we can express this in the form of iterated integrals

> **Fubini's Theorem for Triple Integrals**
>
> If f is continuous on the rectangular box $B = [a, b] \times [c, d] \times [e, f]$, then
>
> $$\iiint\limits_{R} f(x, y, z)\,\mathrm{d}V = \int_{e}^{f} \int_{c}^{d} \int_{a}^{b} f(x, y, z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$$

We may similarly extend this to a triple integral of a function f over a general bounded region E in 3d space (i.e. a solid), nearly identically to how it is done in 2d. The derivation is left as an exercise to the reader.

> **Triple Integrals over General Regions**
>
> For a function f bounded by a region E such that
>
> $$E = \{(x, y, z) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x), u_1(x, y) \leq z \leq u_2(x, y)\}$$
> $$\iiint\limits_{E} f(x, y, z)\,\mathrm{d}V = \int_{a}^{b} \int_{g_1(x)}^{g_2(x)} \int_{h_1(x,y)}^{h_2(x,y)} f(x, y)\,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x$$

This can similarly be extended to work for y or z as the outermost integral.

Note that the problem may sometimes be made easier by first focusing on reducing the problem to 2d, typically by projecting the solid to the xy, yz, or xz plane, depending on the solid type.

For instance, a type 2 region E (as it is termed in the textbook) is defined by

$$E = \{(x, y, z) \mid (y, z) \in D, g_1(y, z) \leq x \leq g_2(y, z)\}$$

Where D is the projection of E onto the yz plane. Then,

$$\iiint\limits_{E} f(x, y, z)\,\mathrm{d}V = \iint\limits_{D} \left[\int_{g_1(x,y)}^{g_2(x,y)} f(x, y, z)\,\mathrm{d}x \right] \mathrm{d}A$$

This can be extended to work for projections onto the xy and xz planes (type 1 and 3, respectively).

We can modify the applications discussed in 15.4 for triple integrals.

$$m = \iiint\limits_E \rho(x,y,z)\,\mathrm{d}V$$

$$M_{yz} = \iiint\limits_E x\rho(x,y,z)\,\mathrm{d}V \qquad M_{xz} = \iiint\limits_E y\rho(x,y,z)\,\mathrm{d}V \qquad M_{xy} = \iiint\limits_E z\rho(x,y,z)\,\mathrm{d}V$$

$$\overline{x} = \frac{M_{yz}}{m} \qquad \overline{y} = \frac{M_{xz}}{m} \qquad \overline{z} = \frac{M_{xy}}{m}$$

$$I_x = \iiint\limits_E (y^2+z^2)\rho(x,y,z)\,\mathrm{d}V \qquad I_y = \iiint\limits_E (x^2+z^2)\rho(x,y,z)\,\mathrm{d}V \qquad I_z = \iiint\limits_E (x^2+y^2)\rho(x,y,z)\,\mathrm{d}V$$

$$Q = \iiint\limits_E \sigma(x,y,z)\,\mathrm{d}V$$

$$P((X,Y,Z)\in E) = \iiint\limits_E f(x,y,z)\,\mathrm{d}V$$

15.7 Triple Integrals in Cylindrical Coordinates

First, some conversions:

$$\begin{aligned} &= r\cos\theta & y &= r\sin\theta & z &= z \\ r^2 &= x^2+y^2 & \tan\theta &= \frac{y}{x} & z &= z \end{aligned}$$

Now, consider E, a type 1 region whose projection D onto the xy-plane is described in polar coordinates. Suppose f is continuous and

$$E = \{(x,y,z) \mid (x,y)\in D, u_1(x,y) \le z \le u_2(x,y)\}$$

Where D is

$$D = \{(r,\theta) \mid \alpha \le \theta \le \beta, h_1(\theta) \le r \le h_2(\theta)\}$$

We know that

$$\iiint\limits_E f(x,y,z)\,\mathrm{d}V = \iint\limits_D \left[\int_{u_1(x,y)}^{u_2(x,y)} f(x,y,z)\,\mathrm{d}z\right]\mathrm{d}A$$

Thus, we can write the following formula:

> **Triple Integration in Cylindrical Coordinates**
>
> $$\iiint\limits_E f(x,y,z)\,\mathrm{d}V = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r\cos\theta, r\sin\theta)}^{u_2(r\cos\theta, r\sin\theta)} f(r\cos\theta, r\sin\theta, z) r\,\mathrm{d}z\,\mathrm{d}r\,\mathrm{d}\theta$$

15.8 Triple Integrals in Spherical Coordinates

Some quick reminders about spherical coordinates

$$P(\rho,\theta,\phi) \qquad \rho\ge 0 \qquad 0\le\phi\le\pi$$

Now, some conversions

$$\begin{aligned} x &= \rho\sin\phi\cos\theta & y &= \rho\sin\phi\sin\theta & z &= \rho\cos\phi \\ \rho &= \sqrt{x^2+y^2+z^2} & \theta &= \arccos\frac{x}{\rho\sin\phi} & \phi &= \arccos\frac{z}{\rho} \end{aligned}$$

Now, with rectangular coordinates, we began by integrating rectangular prisms for triple integrals. In spherical coordinates, we may consider instead a **spherical wedge** E defined by

$$E = \{(\rho,\theta,\phi) \mid a \le p \le b,\ \alpha\le\theta\le\beta,\ c\le\phi\le d\}$$

The derivation will not be covered in detail here, since it is mostly similar to other derivations we have done. Review the textbook if you are interested.

Eventually, we find

> **Triple Integration in Spherical Coordinates**
>
> $$\iiint_E f(x,y,z)\,\mathrm{d}V = \int_c^d \int_\alpha^\beta \int_a^b f(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)\rho^2\sin\phi\,\mathrm{d}\rho\,\mathrm{d}\theta\,\mathrm{d}\phi$$
>
> Where E is a spherical wedge described as above.

The formula can be extended to include more general spherical regions with

$$E = \{(\rho,\theta,\phi) \mid \alpha \le \theta \le \beta,\ c \le \phi \le d,\ g_1(\theta,\phi) \le \rho \le g_2(\theta,\phi)\}$$

15.9 Change of Variables in Multiple Integrals

In other words, u-substitution for multivariate integrals. Consider the change of variables given by a **transformation** T from the uv-plane to the xy-plane

$$T(u,v) = (x,y)$$

Where x, y are described by

$$x = g(u,v) \qquad y = h(u,v)$$

It is also sometimes denoted

$$x = x(u,v) \qquad y = y(u,v)$$

It is typically assumed that T is a C^1 **transformation**, which means that g and h have continuous first-order partial derivatives.

This transformation T is defined as a function whose domain and range are subsets of $\mathbb{R}^2$.

- If $T(u,v) = (x,y)$, the point (x,y) is the **image** of (u,v) for T.
- If $T(u_1,v_1) = T(u_2,v_2) \implies (u_1,v_1) \neq (u_2,v_2)$, i.e. not two points have the same image, T is **injective**.
- If T is injective, $\exists T^{-1}$, the unique inverse transformation from the xy-plane to the uv-plane.

The equations describing the relationship of x, y with u, v can typically be solved in terms of the set of variables describing the image to produce the image under transformation T.

Consider a rectangle S in the uv-plane whose lower left corner is (u_0, v_0) and has dimensions Δu and Δv. Consider that the image of S under transformation T of the region R can be described by the vector

$$\mathbf{r}(u,v) = g(u,v)\mathbf{i} + h(u,v)\mathbf{j}$$

As u, v range over the specified domain. Consider that

$$\mathbf{r}_u = g_u(u_0,v_0)\mathbf{i} + h_u(u_0,v_0)\mathbf{j} = \frac{\partial x}{\partial u}\mathbf{i} + \frac{\partial y}{\partial u}\mathbf{j}$$

And

$$\mathbf{r}_v = g_v(u_0, v_0)\mathbf{i} + h_v(u_0, v_0)\mathbf{j} = \frac{\partial x}{\partial v}\mathbf{i} + \frac{\partial y}{\partial v}\mathbf{j}$$

These are the **tangent vectors** at (x_0, y_0) to the image curve in the xy-plane. The image region $R = T(S)$ can thus be approximated by a parallelogram whose sides are determined by the secant vectors

$$\mathbf{a} = \mathbf{r}(u_0 + \Delta u, v_0) - \mathbf{r}(u_0, v_0) \qquad \mathbf{b} = \mathbf{r}(u_0, v_0 + \Delta v) - \mathbf{r}(u_0, v_0)$$

Consider that

$$\mathbf{r}_u = \lim_{\Delta u \to 0} \frac{\mathbf{r}(u_0 + \Delta u), v_0 - \mathbf{r}(u_0, v_0)}{\Delta u}$$

And similarly for $\mathbf{r}_v$. Thus,

$$\mathbf{a} \approx \Delta u\, \mathbf{r}_u \qquad \mathbf{b} \approx \Delta v\, \mathbf{r}_v$$

Hence, the area of R can be approximated by

$$|(\Delta u\, \mathbf{r}_u) \times (\Delta v\, \mathbf{r}_v)| = \boxed{|\mathbf{r}_u \times \mathbf{r}_v|\, \Delta u\, \Delta v}$$

Note that

$$\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} \mathbf{k}$$

The determinant here is commonly denoted as the *Jacobian*.

The Jacobian of a transformation T given by $x = g(u, v)$ and $y = h(u, v)$ is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}$$

Thus, the area of R, ΔA, can be approximated as

$$\Delta A \approx \left|\frac{\partial(x, y)}{\partial(u, v)}\right| \Delta u\, \Delta v$$

Where the Jacobian is evaluated at (u_0, v_0).

Thus, we come to the topic at hand: changing variables of a double integral.

Change of Variables in a Double Integral

Suppose that T is a C^1 transformation whose Jacobian is nonzero that T maps a region S in the uv-plane onto a region R in the xy-plane. Suppose that f is continuous on R and that R and S are type I or type II plane regions. Suppose also that T is injective, except perhaps on the boundary of S. Then

$$\iint_R f(x, y)\, \mathrm{d}A = \iint_S f(x(u, v), y(u, v)) \left|\frac{\partial(x, y)}{\partial(u, v)}\right| \mathrm{d}u\, \mathrm{d}v$$

As an exercise, consider the transformation T that maps from polar coordinates to rectangle coordinates, i.e. $x = g(r, \theta) = r\cos\theta$ and $y = h(r, \theta) = r\sin\theta$. Show that we derive the same formula for double integration over polar coordinates.

What about triple integrals? We can define the Jacobian of T in 3 dimensions similarly.

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

The formula for changing variables for triple integrals follows similarly. As an exercise, verify this for triple integrals in spherical or cylindrical coordinates.

Chapter 16: Vector Calculus

16.1 Vector Fields

Vector Field

Let D be a set in $\mathbb{R}^2$ (a plane region). A **vector field** on $\mathbb{R}^2$ is a function $\mathbf{F}$ that assigns to each point (x, y) in D a two-dimensional vector $\mathbf{F}(x, y)$.

We can also write $\mathbf{F}$ as

$$\begin{aligned} \mathbf{F}(x, y) = P(x, y)\,\mathbf{i} + Q(x, y)\,\mathbf{j} \\ \mathbf{F} = P\,\mathbf{i} + Q\,\mathbf{j} \end{aligned}$$

Where P, Q are scalar functions of two variables and are sometimes called **scalar fields**.

We can similarly define a vector field on $\mathbb{R}^3, \mathbb{R}^4, \ldots$

16.2 Line Integrals

Consider a curve C given by

$$x = x(t) \qquad y = y(t) \qquad a \leq t \leq b$$

Or, equivalently,

$$\mathbf{r}(t) = x(t)\,\mathbf{i} + y(t)\,\mathbf{j}$$

Assume C is a smooth curve, i.e. $\mathbf{r}'$ is continuous and $\mathbf{r}'(t) = 0$. Then, the line integral of a function $f(x, y)$ along C can be written as the limit of a Riemann sum across the interval $[a, b]$.

$$\int_C f(x, y)\,\mathrm{d}s = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i, y_i)\,\Delta s_i$$

Previously, we found that the arc length of a curve C is

$$L = \int_a^b \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2}\,\mathrm{d}t$$

In other words (by deriving both sides with respect to t),

$$\mathrm{d}s = \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2}\,\mathrm{d}t$$

And thus,

Line Integral

$$\int\limits_C f(x,y)\,\mathrm{d}s = \int_a^b f(x(t),y(t))\sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2}\,\mathrm{d}t$$

Of course, the line integral does not depend on the choice of parametrization of the curve, as long as the curve is traversed exactly once.

Consider now a **piecewise-smooth curve**, i.e. C is essentially several smooth curves continuously and smoothly connected together. Then the line integral on C is simply the sum of the line integrals of each individual smooth sub-curve.

We may also take line integrals of a function $f(x,y)$ along C with respect to either x or y.

$$\int\limits_C f(x,y)\,\mathrm{d}x = \int_a^b f(x(t),y(t))x'(t)\,\mathrm{d}t$$
$$\int\limits_C f(x,y)\,\mathrm{d}y = \int_a^b f(x(t),y(t))y'(t)\,\mathrm{d}t$$

Frequently, line integrals with respect to x and y are summed. This is abbreviated as follows:

$$\int\limits_C f(x,y)\,\mathrm{d}x + \int\limits_C g(x,y)\,\mathrm{d}y = \int\limits_C f(x,y)\,\mathrm{d}x + g(x,y)\,\mathrm{d}y$$

Vector Equation of a Line Segment

As a reminder, the vector equation for a line segment that starts at $\mathbf{r}_0$ and ends at $\mathbf{r}_1$ is given by

$$\mathbf{r}(t) = (1-t)\mathbf{r_0} + t\mathbf{r}_1, \qquad 0 \le t \le 1$$

Note also that a given parametrization determines the **orientation** of a curve C (basically, direction). Thus, $-C$ denotes the curve that is identical to C but has opposite orientation. Then,

$$\int\limits_{-C} f(x,y)\,\mathrm{d}x = -\int\limits_C f(x,y)\,\mathrm{d}x \qquad \int\limits_{-C} f(x,y)\,\mathrm{d}y = -\int\limits_C f(x,y)\,\mathrm{d}y$$

But, if one takes the line integral with respect to arc length, the value of the line integral does *not* change, i.e.

$$\int\limits_{-C} f(x,y)\,\mathrm{d}s = \int\limits_C f(x,y)\,\mathrm{d}s$$

Because, of course, distance is positive.

We may similarly extend 2 dimensional line integrals into 3 dimensional space. In general, we can write the line integral with respect to arc length in n-dimensional space as follows:

$$\int_C f(\mathbf{r}(t))\,\mathrm{d}s = \int_a^b f(\mathbf{r}(t))|\mathbf{r}'(t)|\,\mathrm{d}t$$

And for each variable x_i for $1 \leq i \leq n$,

$$\int_C f(\mathbf{r}(t))\,\mathrm{d}x_i = \int_a^b f(\mathbf{r}(t))x_i'(t)\,\mathrm{d}t$$

Now, consider a continuous vector field $\mathbf{F}$ defined on a smooth curve C. Let $\mathbf{F}$ be given by a vector function $r(t),\ a \leq t \leq b$. Then the line integral of $\mathbf{F}$ along C is

$$\int_C \mathbf{F}\cdot\,\mathrm{d}r = \int_a^b \mathbf{F}(r(t))\cdot\mathbf{r}'(t)\,\mathrm{d}t = \int_C \mathbf{F}\cdot\mathbf{T}\,\mathrm{d}s$$

Where $\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$, which you may recall from the curvature section of 13.3. This integral is commonly used to calculate the work W done by a force vector field $\mathbf{F}$, i.e.

$$W = \int_C \mathbf{F}\cdot\mathbf{T}\,\mathrm{d}s$$

In general, we can say

> Info
>
> $$\int_C F\cdot\,\mathrm{d}\mathbf{r} = \int_C P\,\mathrm{d}x + Q\,\mathrm{d}y + R\,\mathrm{d}z$$
>
> Where $F = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$

16.3 The Fundamental Theorem for Line Integrals

Recall the **Fundamental Theorem of Calculus**:

$$\int_a^b F'(x)\,\mathrm{d}x = F(b) - F(a)$$

Where F' is continuous on the interval $[a, b]$. We can similarly define the **Fundamental Theorem for Line Integrals**:

> Fundamental Theorem for Line Integrals
>
> Let C be a smooth curve given by the vector function $\mathbf{r}(t)$, $a \leq t \leq b$. Let f be a differentiable function of two or three variables whose gradient vector ∇f is continuous on C. Then
>
> $$\int_C \nabla f\cdot\,\mathrm{d}\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

Now, consider two piecewise-smooth curves (commonly denoted **paths**) C_1, C_2 with the same initial point A and terminal point B. Then, by the fundamental theorem

$$\int_{C_1} \nabla f\cdot\,\mathrm{d}\mathbf{r} = \int_{C_2} \nabla f\cdot\,\mathrm{d}\mathbf{r}$$

Whenever ∇f is continuous. That is, the line integral of a **conservative vector field** depends only on the initial point and terminal point of a curve.

If F is a continuous vector field with domain D, the line integral $\int_C F \cdot d\mathbf{r}$ is **independent of path** if the above is satisfied for any two paths C_1 and C_2.

> **Closed Curve**
>
> A closed curve is a curve whose terminal point coincides with its initial point. Let A, B be points on a closed curve C. Let C_1 be the curve from $A \to B$ and C_2 be the curve $B \to A$. Then
>
> $$\int_C F \cdot d\mathbf{r} = \int_{C_1} F \cdot d\mathbf{r} + \int_{C_2} F \cdot d\mathbf{r} = \int_{C_1} F \cdot d\mathbf{r} - \int_{-C_2} F \cdot d\mathbf{r} = 0$$

The following theorem can be trivially shown.

> **Info**
>
> $\int_C F \cdot d\mathbf{r}$ is independent of path in D if and only if $\int_C F \cdot d\mathbf{r} = 0$ for every closed path C in D.

Let an **open domain** D be a domain such for every point $P \in D$, there is a disk with center P that lies entirely in D. Additionally, let a **connected** domain D be a domain such that any two points in D can be connected via a path that lies entirely in D.

> **Info**
>
> Suppose F is a vector field that is continuous on an open connected region D. If $\int_C F \cdot d\mathbf{r}$ is independent of path in D, then F is a conservative vector field on D, i.e. there exists a function f such that $\nabla f = F$.

See the textbook for a proof of this theorem.

We do, however, still need to determine if a vector field F is conservative. Suppose $F = P\mathbf{i} + Q\mathbf{j}$ is conservative, where P, Q have continuous first-order partial derivatives. Then there exists a function f such that $F = \nabla f$ such that

$$P = \frac{\partial f}{\partial x} \qquad Q = \frac{\partial f}{\partial y}$$

By Clairaut's Theorem

$$\frac{\partial P}{\partial y} = \frac{\partial^2 f}{\partial y\, \partial x} = \frac{\partial^2 f}{\partial x\, \partial y} = \frac{\partial Q}{\partial x}$$

Thus,

> **Info**

If $F(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$ is a conservative vector field, where P, Q have continuous first-order partial derivatives on a domain D, then throughout D it must be true that $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

The converse of this is only true for certain regions. Let a **simple curve** denote a curve that doesn't intersect itself anywhere between its endpoints. Then, let a **simply-connected region** D be a region such that every simple closed curve in D encloses only points that are in D. Intuitively, this means D contains no holes and is one contiguous region. Then,

Info

Let $F = P\mathbf{i} + Q\mathbf{j}$ be a vector field on an open simply-connected region D. Suppose P, Q have continuous first-order partial derivatives and $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ throughout D. Then F is conservative.

The ideas derived in this chapter can be applied to a continuous force field F (i.e. physics).

Using the fact that $F(\mathbf{r}(t)) = m\mathbf{r}''(t)$, where $\mathbf{r}(t)$ is the position function (this is the familiar equation $F = ma$), we can derive, from endpoints $a \to b$.

$$\begin{aligned} W &= \frac{1}{2}m|\mathbf{r}'(b)|^2 - \frac{1}{2}m|\mathbf{r}'(a)|^2 \\ &= K(B) - K(A) \end{aligned}$$

Or, in other words, the change in kinetic energy of the object.

Meanwhile, if we assume F is a conservative force field, i.e. we can write $F = \nabla f$ for some function f, we can use the equation $P(x, y, z) = -f(x, y, z)$, where P is the potential energy function. This naturally leads to $F = -\nabla P$, from which we easily derive, from endpoints $a \to b$.

$$\begin{aligned} W &= P(\mathbf{r}(a)) - P(\mathbf{r}(b)) \\ &= P(A) - P(B) \end{aligned}$$

Thus,

$$\begin{aligned} K(B) - K(A) &= P(A) - P(B) \\ P(A) + K(A) &= P(B) + K(B) \end{aligned}$$

In other words, we just derived the **Law of Conservation of Energy**!

16.4 Green's Theorem

Green's Theorem helps relate a line integral around a simple closed curve C and a double integral over the plane region D bounded by C (the interior of C, essentially).

Warning

For these notes, we will follow the convention of the textbook, i.e. the **positive orientation** of C refers to a single **counterclockwise** traversal of C.

Green's Theorem

Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C. If P, Q have continuous partial derivatives on an open on an open region that contains D, then

$$\int\limits_C P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint\limits_D \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\,\mathrm{d}A = \int\limits_{\partial D} P\,\mathrm{d}x + Q\,\mathrm{d}y$$

Positive Orientation Line Integrals

The expression

$$\oint\limits_C P\,\mathrm{d}x + Q\,\mathrm{d}y$$

also denotes a line integral on curve C calculated using the positive orientation.

Green's theorem is essentially the analogue of the Fundamental Theorem of Calculus for double integrals.

The proof is left out for brevity. As food for thought, consider that, to prove Green's Theorem when D is a simple region, it's only necessary to show

$$\int\limits_C P\,\mathrm{d}x = -\iint\limits_D \frac{\partial P}{\partial y}\,\mathrm{d}A \qquad \int\limits_C Q\,\mathrm{d}y = \iint\limits_D \frac{\partial Q}{\partial x}\,\mathrm{d}A$$

Then, consider that proving Green's Theorem for a nonsimple region D that is a finite union of simple regions is equivalent to dividing D into simple regions and consider each simple region separately. (Hint: you will get some cancellation integrating along the boundary of the two simple curves).

Tip

Green's Theorem can be used in both ways. Sometimes, you convert from a single integral to a double integral. And other times, you convert from a double integral to a single integral. The book has many examples to consider.

For instance, consider trying to compute the area of a region D. We have that $A(D) = \iint\limits_D 1\,\mathrm{d}A \implies \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$. There are many different possibilities for $P(x, y)$ and $Q(x, y)$ for this to be true. In general, we can write, by Green's Theorem,

$$A = \oint\limits_C x\,\mathrm{d}y = -\oint\limits_C y\,\mathrm{d}x = \frac{1}{2}\oint\limits_C x\,\mathrm{d}y - y\,\mathrm{d}x$$

Figuring out the values of $P(x, y)$ and $Q(x, y)$ that produce this formula is left as an exercise to the reader.

Consider the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Find the area enclosed by the ellipse.

Parametrically, $x = a\cos t$ and $y = b\sin t$, with $0 \leq t \leq 2\pi$. We can then write

$$\begin{aligned} A &= \frac{1}{2}\oint_C x\,\mathrm{d}y - y\,\mathrm{d}x \\ &= \frac{1}{2}\int_0^{2\pi} (a\cos t)(b\cos t)\,\mathrm{d}t - (b\sin t)(-a\sin t)\,\mathrm{d}t \\ &= \frac{ab}{2}\int_0^{2\pi} \mathrm{d}t \\ &= \boxed{\pi ab} \end{aligned}$$

Recall the ideas given for the proof of Green's Theorem for nonsimple regions that are the finite union of simple regions. This idea frequently be applied to solve problems for such regions using Green's Theorem (i.e., dividing the nonsimple region into it simple regions and applying Green's Theorem individually).

Example

Consider two curves C and C'. Let $D(C)$ denote the region enclosed by C, and analogously for C'. Let $D(C) \subset D(C')$. Then, $D(C - C')$ is a nonsimple region with a hole.

The positively oriented boundary of this region is $C \cap (-C')$. (This follows from $C - C'$). Then, by Green's Theorem, we can write

$$\int_C P\,\mathrm{d}x + Q\,\mathrm{d}y + \int_{-C'} P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint_D \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\,\mathrm{d}A$$

16.5 Curl and Divergence

Curl

We begin with a discussion of the property of a vector field known as the **curl**. The curl vector is named as such because it describes the rotational motion of particles. Particles at (x, y, z) in this vector field will tend to rotate around the axis aligned with the direction of the curl, and at a rate proportional to the curl vector's magnitude.

If $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ is a vector field on $\mathbb{R}^3$ and the partial derivatives of P, Q, R all exist, then the curl of $\mathbf{F}$ is the vector field on $\mathbb{R}^3$ defined by

$$\text{curl } \mathbf{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathbf{k}$$

It is easier to remember this using operator notation. We define the following as the **vector differential operator** ∇:

$$\nabla = \mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}$$

We have seen this before in the notation of the gradient of some vector function f:

$$\nabla f = \mathbf{i}\frac{\partial f}{\partial x} + \mathbf{j}\frac{\partial f}{\partial y} + \mathbf{k}\frac{\partial f}{\partial z}$$

Knowing this, we may now write

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \begin{matrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{matrix}$$

There are a few nice facts about the curl of a vector field.

Curl Properties

1. If $f(x, y, z)$ has continuous second-order partial derivatives, then $\operatorname{curl}(\nabla f)=\mathbf{0}$.
2. If $\mathbf{F}$ is a conservative vector field, by definition, $\mathbf{F}=\nabla f$. Thus, a conservative vector field $\mathbf{F}$ has curl $\mathbf{F}=0$.
3. If $\mathbf{F}$ is a vector field defined on all of $\mathbb{R}^3$ whose component functions have continuous partial derivatives and curl $\mathbf{F}=\mathbf{0}$, then $\mathbf{F}$ is a conservative vector field.

The proof of property 1 is easy, and is left as an exercise to the reader. The proof of property 3 require topics we have yet to discuss—in particular, Stokes' Theorem—and so will not be shown.

Note also the following ideas:

Tip

- If $\operatorname{curl} \mathbf{F}=\mathbf{0}$ at a point P, then the fluid is free from rotations at P and $\mathbf{F}$ is called **irrotational** at P. Note that particles still move with this fluid, but do not rotate around its axis.
- If $\operatorname{curl} \mathbf{F} \neq \mathbf{0}$, then particles do in fact rotate about its axis.

Divergence

The **divergence** of a vector field is again related to fluid dynamics. In particular, if $\mathbf{F}(x, y, z)$ is the velocity of a fluid, then its divergence is the net rate of change with respect to time of the mass of fluid flowing from point (x, y, z) per unit volume. That is, it measures the fluid's tendency to diverge from the point.

If $\mathbf{F}=P\mathbf{i}+Q\mathbf{j}+R\mathbf{k}$ is a vector field on $\mathbb{R}^3$ and $\frac{\partial P}{\partial x}, \frac{\partial Q}{\partial y}, \frac{\partial R}{\partial z}$ all exist, then the divergence of $\mathbf{F}$ is described by

$$\operatorname{div} \mathbf{F}=\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}$$

Note that the divergence of $\mathbf{F}$ is in fact a scalar field, and not a vector field like curl $\mathbf{F}$.

We can also represent the divergence of F in operator notation.

$$\operatorname{div} \mathbf{F}=\nabla \cdot \mathbf{F}$$

Since the curl of $\mathbf{F}$ is also a vector field on $\mathbb{R}^3$ if $\mathbf{F}$ is a vector field on $\mathbb{R}^3$, we can compute the divergence of curl $\mathbf{F}$ as well. In fact,

Info

If $\mathbf{F}=P\mathbf{i}+Q\mathbf{j}+R\mathbf{k}$ is a vector field on $\mathbb{R}^3$ and P, Q, R have continuous second-order partial derivatives, then $\operatorname{div} \operatorname{curl} \mathbf{F}=0$.

The proof of this is fairly straightforward, and is left as an exercise to the reader.

Note also the following ideas:

Tip

- If $\operatorname{div} \mathbf{F}=0$, $\mathbf{F}$ is said to be **incompressible**.

We may also compute the divergence of a gradient vector field.

$$\text{div}\,(\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

This expression occurs frequently, and is thus abbreviated as $\nabla^2 f$. It is also denoted as the **Laplace operator** because of its relation to *Laplace's Equation*:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

Vector Forms of Green's Theorem

Using the curl and divergence operators, we can rewrite Green's Theorem in vector form.

Curl

Consider the plane region D, its boundary curve C, and the functions P, Q that satisfy the previously stated hypotheses of Green's Theorem. Consider also the vector field $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$. Then, its line integral is

$$\oint_C \mathbf{F} \cdot \,\mathrm{d}\mathbf{r} = \oint_C P\,\mathrm{d}x + Q\,\mathrm{d}y$$

Considering $\mathbf{F}$ as a vector field on $\mathbb{R}^3$ with third component 0, we derive

$$\text{curl}\,\mathbf{F} = \begin{matrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P(x,y) & Q(x,y) & 0 \end{matrix} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathbf{k}$$

That is,

$$(\text{curl}\,\mathbf{F}) \cdot k = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

Which allows us to express Green's Theorem in vector form:

> Info
>
> $$\oint_C \mathbf{F}\,\mathrm{d}\mathbf{r} = \iint_D (\text{curl}\,\mathbf{F}) \cdot \mathbf{k}\,\mathrm{d}A$$

That is, the line integral of the tangential component of $\mathbf{F}$ along C is equivalent to the double integral of the vertical component of curl $\mathbf{F}$ over the region D enclosed by C.

Divergence

We can derive a similar formula involving the *normal* component of $\mathbf{F}$.

Let C be given by

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}, \quad a \le t \le b$$

Then, we can write expressions for the unit tangent vector

$$\mathbf{T}(t) = \frac{x'(t)}{|\mathbf{r}'(t)|}\mathbf{i} + \frac{y'(t)}{|\mathbf{r}'(t)|}\mathbf{j}$$

and the outward unit normal vector

$$\mathbf{n}(t) = \frac{y'(t)}{|\mathbf{r}'(t)|}\mathbf{i} - \frac{x'(t)}{|\mathbf{r}'(t)|}\mathbf{j}$$

It follows that

$$\begin{aligned}
\oint_C \mathbf{F}\cdot\mathbf{n}\,\mathrm{d}s &= \int_a^b (\mathbf{F}\cdot\mathbf{n})(t)|\mathbf{r}'(t)|\,\mathrm{d}t \\
&= \int_a^b \left[\frac{P(x(t),y(t))y'(t)}{|\mathbf{r}'(t)|} - \frac{Q(x(t),y(t))x'(t)}{|\mathbf{r}'(t)|}\right]|\mathbf{r}'(t)|\,\mathrm{d}t \\
&= \int_a^b P(x(t),y(t))y'(t)\,\mathrm{d}t - Q(x(t),y(t))x'(t)\,\mathrm{d}t \\
&= \int_C P\,\mathrm{d}y - Q\,\mathrm{d}x = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right)\mathrm{d}A \\
&= \iint_D \operatorname{div}\mathbf{F}(x,y)\,\mathrm{d}A
\end{aligned}$$

Thus,

Info

$$\oint_C \mathbf{F}\cdot\mathbf{n}\,\mathrm{d}s = \iint_D \operatorname{div}\mathbf{F}(x,y)\,\mathrm{d}A$$

That is, the line integral of the normal component of $\mathbf{F}$ along C is equal to the double integral of the divergence of $\mathbf{F}$ over the region D enclosed by C.

16.6 Parametric Surfaces and Their Areas

Parametric surfaces are the analogues of parametric functions in 3 dimensions. They can be described by a vector function $\mathbf{r}(u,v)$:

$$\mathbf{r}(u,v) = x(u,v)\mathbf{i} + y(u,v)\mathbf{j} + z(u,v)\mathbf{k}$$

This is a vector-valued function defined on a region D in the uv-plane, and the **parametric surface** S is created by (u,v) varying throughout D. The equations

$$x = x(u,v) \qquad y = y(u,v) \qquad z = z(u,v)$$

Are known as the **parametric equations** of S.

For a parametric surface S, there are two useful families of curves that lie on S: constant u and constant v, i.e. the curves that correspond to the vertical and horizontal lines in the uv-plane. These curves are known as **grid curves**, which naturally follows from the curves' display in the uv-plane.

Surfaces of revolution can be expressed parametrically, e.g. the surface generated by rotating a curve $y = f(x)$ around the x-axis. Let θ be the angle of rotation of some point on the surface. Then, we can describe a point (x,y,z) on S as

$$x = x \qquad y = f(x)\cos\theta \qquad z = f(x)\sin\theta$$

It may help to verify this for yourself with an example function $f(x)$.

We can also describe the tangent planes to a parametric surface S. Let S be traced out by the vector function

$$\mathbf{r}(u,v) = x(u,v)\mathbf{i} + y(u,v)\mathbf{j} + z(u,v)\mathbf{k}$$

at a point P_0 with position vector $\mathbf{r}(u_0,v_0)$. Now, consider the grid curve C_1 lying on S that occurs when u is constant, i.e. $u = u_0$. Then, the tangent vector to C_1 at P_0 is obtained by taking the partial derivative of $\mathbf{r}$ with respect to v:

$$\mathbf{r}_v = \frac{\partial x}{\partial v}(u_0, v_0)\mathbf{i} + \frac{\partial y}{\partial v}(u_0, v_0)\mathbf{j} + \frac{\partial z}{\partial v}(u_0, v_0)\mathbf{k}$$

Similarly, when v is kept constant, i.e. $v = v_0$, then the tangent vector to the grid curve C_2 at P_0 is

$$\mathbf{r}_u = \frac{\partial x}{\partial u}(u_0, v_0)\mathbf{i} + \frac{\partial y}{\partial u}(u_0, v_0)\mathbf{j} + \frac{\partial z}{\partial u}(u_0, v_0)\mathbf{k}$$

Info

If $\mathbf{r}_u \times \mathbf{r}_v \neq \mathbf{0}$, then the surface S is **smooth**.

For a smooth surface, the **tangent plane** is the plane that contains these two tangent vectors, and $\mathbf{r}_u \times \mathbf{r}_v$ is a normal vector to the tangent plane.

Finally, we can consider calculating the *surface area* of a general parametric surface. The derivation is essentially a Riemann sum over the surface areas of sections of the tangent plane approximations of the surface, and thus will be skipped. The following theorem describes the formula succinctly:

Info

Let a smooth parametric surface S be given by

$$\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}, \qquad (u, v) \in D$$

Let S be covered only once as (u, v) range throughout D (no overcounting/overlapping!). Then, the surface area of S is simply

$$A(S) = \iint_D |\mathbf{r}_u \times \mathbf{r}_v| \, \mathrm{d}A$$

where

$$\mathbf{r}_u = \frac{\partial x}{\partial u}\mathbf{i} + \frac{\partial y}{\partial u}\mathbf{j} + \frac{\partial z}{\partial u}\mathbf{k} \qquad \mathbf{r}_v = \frac{\partial x}{\partial v}\mathbf{i} + \frac{\partial y}{\partial v}\mathbf{j} + \frac{\partial z}{\partial v}\mathbf{k}$$

Note that when we have $z = f(x, y)$, we can easily derive the surface area formula found in 15.5 Surface Area by considering the parametrization $x = x, \ y = y, \ z = f(x, y)$. That is,

$$A(S) = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} \, \mathrm{d}A$$

16.7 Surface Integrals

Surface integrals are the the line integral equivalent in 3 dimensions. That is, surface integrals are to surface area as line integrals are to arc length.

We will soon define the surface integral of the function $f(x, y, z)$ over the surface S. Note that when $f(x, y, z) = 1$, this is equivalent to the surface area of S. (Much like how the line integral of $f(x, y) = 1$ is simply the arc length). We begin with parametric surfaces.

Parametric Surfaces

Suppose S is described by

$$\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}, \qquad (u, v) \in D$$

Again, we derive the formula for the surface integral via a Riemann sum. In this case, over infinitesimal subrectangles of the surface (specifically, taken from the tangent plane at each point in the surface), i.e.

$$\iint\limits_S f(x,y,z)\,\mathrm{d}S = \lim_{m,n\to\infty} \sum_{i=1}^{m}\sum_{j=1}^{n} f(P_{ij})\,\Delta S_{ij}$$

Where P_{ij} is the evaluation of f at some point within the subrectangle and ΔS_{ij} is the area of the the subrectangle with indices (i,j). Then, since ΔS_{ij} may be approximated by the area of a parallelogram in the tangent plane, i.e. $\Delta S_{ij} \approx |\mathbf{r}_u \times \mathbf{r}_v|\,\Delta u\,\Delta v$, where the tangent vectors at the point P_{ij} are described by $\mathbf{r}_u = \frac{\partial x}{\partial u}\mathbf{i} + \frac{\partial y}{\partial u}\mathbf{j} + \frac{\partial z}{\partial u}\mathbf{k}$ and $\mathbf{r}_v = \frac{\partial x}{\partial v}\mathbf{i} + \frac{\partial y}{\partial v}\mathbf{j} + \frac{\partial z}{\partial v}\mathbf{k}$. Thus,

> **Parametric Surface Integral**
>
> $$\iint\limits_S f(x,y,z)\,\mathrm{d}S = \iint\limits_D f(\mathbf{r}(u,v))|\mathbf{r}_u \times \mathbf{r}_v|\,\mathrm{dA}$$

Note the similarity to the line integral formula: $\int\limits_C f(x,y,z)\,\mathrm{d}s = \int_a^b f(\mathbf{r}(t))|\mathbf{r}'(t)|\,\mathrm{d}t$.

Also note that if S is a piecewise-smooth surface that intersect only on their boundaries, then it naturally follows that

$$\iint\limits_S f(x,y,z)\,\mathrm{d}S = \sum_{i=1}^{n}\left[\iint\limits_{S_1} f(x,y,z)\,\mathrm{d}S\right]$$

Surface integrals may be applied to the same real-world ideas we have previously discussed. For instance, the total mass of a surface S whose density is described by $\rho(x,y,z)$ is

$$m = \iint\limits_S \rho(x,y,z)\,\mathrm{d}S$$

And the center of mass is $(\overline{x}, \overline{y}, \overline{z})$, where

$$\overline{x} = \frac{1}{m}\iint\limits_S x\rho(x,y,z)\,\mathrm{d}S \qquad \overline{y} = \frac{1}{m}\iint\limits_S y\rho(x,y,z)\,\mathrm{d}S \qquad \overline{z} = \frac{1}{m}\iint\limits_S z\rho(x,y,z)\,\mathrm{d}S$$

Moments of inertia, etc. can similarly be derived for surfaces.

Graphs of Functions

Now, consider a surface S with equation $z = g(x,y)$. Then, it is a parametric surface with the following equations:

$$x = x \qquad y = y \qquad z = g(x,y)$$

Thus,

$$\mathbf{r}_x = \mathbf{i} + \left(\frac{\partial q}{\partial x}\right)\mathbf{k} \qquad \mathbf{r}_y = \mathbf{j} + \left(\frac{\partial g}{\partial y}\right)\mathbf{k}$$

And taking the cross product,

$$\mathbf{r}_x \times \mathbf{r}_y = -\frac{\partial g}{\partial x}\mathbf{i} - \frac{\partial g}{\partial y}\mathbf{j} + \mathbf{k} \implies |\mathbf{r}_x \times \mathbf{r}_y| = \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}$$

Substituting into our previously derived formula, we get

Surface Integral when $z = g(x, y)$

$$\iint_S f(x,y,z)\,\mathrm{d}S = \iint_D f(x,y,g(x,y))\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}\,\mathrm{d}A$$

Oriented Surfaces

Like oriented line integrals, we have oriented surfaces. The purpose is to define surface integrals of vector fields.

Warning

A slight caveat with oriented surfaces, though, is that there do actual exist nonorientable surfaces! Consider, for example, the Möbius strip. Pick any point P, and choose a side of the Möbius strip. Now, walk around the Möbius strip until you reach P again for the first time. If you do it right, you shouldn't end up back on the same side! This is because the Möbius strip really only has one side.

Thus, we will consider only *orientable* (two-sided) surfaces.

Consider a surface S that has a tangent plane at every point (x, y, z) on S, disregarding boundary points. There exist two unit normal vectors $\mathbf{n}_1, \mathbf{n}_2$ with $\mathbf{n}_2 = -\mathbf{n}_1$ at (x, y, z).

Choose one of the two unit normal vectors. Let $\mathbf{n}$ denote this chosen unit vector. Then, if $\mathbf{n}$ varies continuously over the entirety of S, then S is an **oriented surface** and the given choice of $\mathbf{n}$ provides S with an **orientation**. (Also, note that if your initial choice of $\mathbf{n}$ doesn't vary continuously over, the other choice of $\mathbf{n}$ wouldn't work either, since $\mathbf{n}_2 = -\mathbf{n}_1$).

For a surface S described by $z = g(x, y)$, we can describe $\mathbf{n}$ using the cross product of the tangent vectors described previously and repeated below for the reader's convenience.

$$\mathbf{r}_x \times \mathbf{r}_y = -\frac{\partial g}{\partial x}\mathbf{i} - \frac{\partial g}{\partial y}\mathbf{j} + \mathbf{k}$$

This is a vector function that describes the normal vector at every point in the surface. So, it suffices to normalize this expression to find $\mathbf{n}$.

$$\mathbf{n} = \frac{-\frac{\partial g}{\partial x}\mathbf{i} - \frac{\partial g}{\partial y}\mathbf{j} + \mathbf{k}}{\sqrt{1 + \left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2}}$$

Note that the *upward* orientation of the surface is given by this expression since the $\mathbf{k}$-component is positive, and the *downward* orientation is given by $-\mathbf{n}$.

And if S is a smooth orientable surface given parametrically by the vector function $\mathbf{r}(u, v)$, then it is simply

$$\mathbf{n} = \frac{\mathbf{r}_u \times \mathbf{r}_v}{|\mathbf{r}_u \times \mathbf{r}_v|}$$

Surface Integrals of Vector Fields

Consider an oriented surface S with unit normal vector $\mathbf{n}$. Now, let there exist a fluid with density $\rho(x, y, z)$ and velocity field $\mathbf{v}(x, y, z)$ that is flowing through S. The rate of flow per unit area may be described by $\rho\mathbf{v}$.

If we consider infinitesimal subsections of S denoted S_{ij}, then S_{ij} is approximately planar. Thus, the mass of fluid per unit time crossing S_{ij} in the direction of the unit normal vector $\mathbf{n}$ is

$$(\rho\mathbf{v} \cdot \mathbf{n})(A(S_{ij}))$$

Where $A(S_{ij})$ denotes the area of S_{ij} and $\rho, \mathbf{v}, \mathbf{n}$ are all evaluated at some point on S_{ij}. (For clarity, consider that $\rho\mathbf{v} \cdot \mathbf{n}$ is the magnitude of the component of the vector $\rho\mathbf{v}$ in the direction of the unit vector $\mathbf{n}$). By performing a Riemann sum over these infinitesimal S_{ij}'s, we get

$$\iint\limits_S \rho\mathbf{v} \cdot \mathbf{n}\,\mathrm{d}S = \iint\limits_S \rho(x,y,z)\mathbf{v}(x,y,z) \cdot \mathbf{n}(x,y,z)\,\mathrm{d}S$$

And if we write $\mathbf{F} = \rho\mathbf{v}$, then $\mathbf{F}$ is a vector field on $\mathbb{R}^3$ and the integral becomes

$$\iint\limits_S \mathbf{F} \cdot \mathbf{n}\,\mathrm{d}S$$

This is a very common integral, particularly in physics, even when $\mathbf{F}$ is not $\rho\mathbf{v}$, and thus is denoted as the **surface integral** or **flux integral** of $\mathbf{F}$ over S. It's also called the **flux** of $\mathbf{F}$ across S.

Flux

If $\mathbf{F}$ is a continuous vector field defined on an oriented surface S with unit normal vector $\mathbf{n}$, then the **surface integral of $\mathbf{F}$ over S** is

$$\iint\limits_S \mathbf{F} \cdot\,\mathrm{d}S = \iint\limits_S \mathbf{F} \cdot \mathbf{n}\,\mathrm{d}S$$

When S is parametrized by a vector function $\mathbf{r}(u,v)$, then

$$\iint\limits_S \mathbf{F} \cdot\,\mathrm{d}S = \iint\limits_S \mathbf{F} \cdot \frac{\mathbf{r}_u \times \mathbf{r}_v}{|\mathbf{r}_u \times \mathbf{r}_v|}\,\mathrm{d}S = \iint\limits_D \mathbf{F}(\mathbf{r}(u,v)) \cdot \frac{\mathbf{r}_u \times \mathbf{r}_v}{|\mathbf{r}_u \times \mathbf{r}_v|} \cdot |\mathbf{r}_u \times \mathbf{r}_v|\,\mathrm{d}A$$

That is,

$$\iint\limits_S \mathbf{F} \cdot\,\mathrm{d}S = \iint\limits_D \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v)\,\mathrm{d}A$$

Meanwhile, if the surface S is given by $z = g(x,y)$, then x, y can be considered the parameters and we can write

$$\mathbf{F} \cdot (\mathbf{r}_x \times \mathbf{r}_y) = (P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}) \cdot \left(-\frac{\partial g}{\partial x}\mathbf{i} - \frac{\partial g}{\partial y}\mathbf{j} + \mathbf{k}\right) = -P\frac{\partial g}{\partial x} - Q\frac{\partial g}{\partial y} + R$$

Thus, we can write

$$\iint\limits_S \mathbf{F} \cdot\,\mathrm{d}S = \iint\limits_D \left(-P\frac{\partial g}{\partial x} - Q\frac{\partial g}{\partial y} + R\right)\mathrm{d}A$$

This formula assumes upward orientation of s. For downward orientation, simply multiply by -1.

Finally, let's consider some applications of the surface integral of a vector field to physics.

Consider an electric field $\mathbf{E}$. Then the surface integral $\iint\limits_S \mathbf{E} \cdot\,\mathrm{d}S$ is called the **electric flux** of $\mathbf{E}$ through the surface S. One important law of electrostatics is **Gauss's Law**, which states that the net charge enclosed by a closed surface S is $Q = \epsilon_0 \iint\limits_S \mathbf{E} \cdot\,\mathrm{d}S$, for some constant ϵ_0 known as the permittivity of free space.

Consider now the scalar field $u(x,y,z)$ that describes the temperature at a point (x,y,z) within a substance. Then the **heat flow** is defined by the vector field $\mathbf{F} = -K\,\nabla u$, where K is the **conductivity** of the substance (and is typically experimentally determined). Then, the rate of heat flow across a surface S in the substance is $\iint\limits_S \mathbf{F} \cdot \mathrm{d}S = -K \iint\limits_S \nabla u \cdot \mathrm{d}S$.

16.8 Stokes' Theorem

It's helpful to think of Stokes' Theorem as a higher-dimensional version of Green's Theorem. Green's Theorem relates a double integral over a plane region D to a line integral around its plane boundary curve. Similarly, Stokes' Theorem relates a surface integral over a surface S to a line integral around the boundary curve of S, a space curve.

In other words, Stokes' Theorem is a more generalized analogue of Green's Theorem.

Like Green's Theorem, Stokes' Theorem also includes the notion of orientation. In particular, as described in the last section, an oriented surface S possesses a unit normal vector $\mathbf{n}$ at every point. (Either upwards or downwards). Then, we define the **positive orientation of the boundary curve** C for a chosen orientation of S. Essentially, if you walk in the positive direction around the C with your head pointing in the direction of $\mathbf{n}$, then the surface S will always be on your left.

> **Stokes' Theorem**
>
> Let S be an oriented piecewise-smooth surface that is bounded by a simple, closed, piecewise-smooth boundary curve C with positive orientation. Let $\mathbf{F}$ be a vector field whose components have continuous partial derivatives on an open region in $\mathbb{R}^3$ that contains S. Then
>
> $$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \iint_S \operatorname{curl} \mathbf{F} \cdot \mathrm{d}\mathbf{S}$$
>
> The positively oriented boundary curve of the oriented surface S is denoted ∂S, so Stokes' Theorem can also be written as
>
> $$\int_{\partial S} \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \iint_S \operatorname{curl} \mathbf{F} \cdot \mathrm{d}\mathbf{S}$$

There are some equivalencies here that should be noted. In particular,

$$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{T}\, \mathrm{d}s \qquad \text{and} \qquad \iint_S \operatorname{curl} \mathbf{F} \cdot \mathrm{d}\mathbf{S} = \iint_S \operatorname{curl} \mathbf{F} \cdot \mathbf{n}\, \mathrm{d}\mathbf{S}$$

Substituting these into Stokes' Theorem reveals that the line integral around the boundary curve of S of the tangential component of $\mathbf{F}$ is equal to the surface integral over S of the normal component of the curl of $\mathbf{F}$.

The existence of Stokes' Theorem as a "generalized Green's Theorem" now becomes clear when S lies entirely in the xy-plane with upward orientation. In this case, the unit normal is $\mathbf{k}$ and the surface integral becomes a double integral of the area in the xy-plane, i.e.

$$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \iint_S \operatorname{curl} \mathbf{F} \cdot \mathrm{d}\mathbf{S} = \iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{k}\, \mathrm{d}A$$

This should look familiar, since it's just the formula from [Vector Forms of Green's Theorem].

Also, note the following key fact about Stokes' Theorem.

> **Tip**
>
> If S_1, S_2 are oriented surfaces with the same oriented boundary curve C and both satisfy the hypotheses of Stokes' Theorem, then
>
> $$\iint_{S_1} \operatorname{curl} \mathbf{F} \cdot \mathrm{d}\mathbf{S} = \int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \iint_{S_2} \operatorname{curl} \mathbf{F} \cdot \mathrm{d}\mathbf{S}$$

This is helpful whenever it's easier to integrate over another surface with the same boundary curve.

Stokes' Theorem is challenging to prove for all pairs of a surface S and a boundary curve C satisfying its requirements, and thus there will be no proof provided. The textbook does include a proof of a special case of Stokes' Theorem; give it a read if you're interested!

Stokes' Theorem enables us to interpret the physical meaning of the curl vector. Let C be an oriented closed curve and $\mathbf{v}$ represent the velocity field in fluid flow. Consider the line integral $\int\limits_C \mathbf{v} \cdot \,\mathrm{d}\mathbf{r} = \int\limits_C \mathbf{v} \cdot \mathbf{T} \,\mathrm{d}s$. Note that $\mathbf{v} \cdot \mathbf{T}$ is the component of $\mathbf{v}$ in the direction of the unit tangent vector $\mathbf{T}$. That is, the closer the directions of $\mathbf{v}$ and $\mathbf{T}$, the larger the value of $\mathbf{v} \cdot \mathbf{T}$. In other words, $\int\limits_C \mathbf{v} \cdot \,\mathrm{d}\mathbf{r}$ is a measure of the tendency of the fluid to move around C in the positive orientation, a quantity known as the **circulation** of $\mathbf{v}$ around C. (When $\int\limits_C \mathbf{v} \cdot \,\mathrm{d}\mathbf{r}$ is negative, that means the fluid tends to circulate around C in the other direction).

We can in fact approximate the circulation at a point with the following equation:

$$\text{curl } \mathbf{v}(P_0) \cdot \mathbf{n}(P_0) = \lim_{a \to 0} \frac{1}{\pi a^2} \int\limits_{C_a} \mathbf{v} \cdot \,\mathrm{d}\mathbf{r}$$

Where C_a is the boundary circle of a small disk S_a with radius a and center P_0. That is, curl $\mathbf{v} \cdot \mathbf{n}$ measures the rotating effect of the fluid about the axis $\mathbf{n}$. And thus the curling effect is greatest about the axis parallel to curl $\mathbf{v}$.

Finally, armed with Stokes' Theorem, we can now (mostly) prove one of the statements we made in 16.5 Curl and Divergence: If $\mathbf{F}$ is a vector field defined on all of $\mathbb{R}^3$ whose component functions have continuous partial derivatives and curl $\mathbf{F} = \mathbf{0}$, then $\mathbf{F}$ is a conservative vector field.

We know that $\mathbf{F}$ is conservative if $\int\limits_C \mathbf{F} \cdot \,\mathrm{d}\mathbf{r} = 0$ for every closed path C. Note that we can guarantee that C bounds an orientable surface S. (However, proving this statement requires more advanced techniques, hence why this is only a mostly complete proof). Assuming we know this is true, though, Stokes' Theorem states

$$\int\limits_C \mathbf{F} \cdot \,\mathrm{d}\mathbf{r} = \iint\limits_S \text{curl } \mathbf{F} \cdot \,\mathrm{d}\mathbf{S} = \iint\limits_S \mathbf{0} \cdot \,\mathrm{d}\mathbf{S} = 0$$

And if C is not simple, it can be divided into multiple simple curves, and the integrals around these simple curves are all 0. Hence, the equality holds for nonsimple curves too.

16.9 The Divergence Theorem

Recall that, in this section of 16.5, we rewrote Green's Theorem in vector form using curl and divergence. In particular, the divergence formula was

$$\oint\limits_C \mathbf{F} \cdot \mathbf{n} \,\mathrm{d}s = \iint\limits_D \text{div } \mathbf{F}(x, y) \,\mathrm{d}A$$

Just like how Stokes' Theorem was essentially an extension to $\mathbb{R}^3$ of the vector form of Green's Theorem using curl, the Divergence Theorem is an extension to $\mathbb{R}^3$ of the other vector form of Green's Theorem displayed above.

In particular, we shall consider regions E that are simultaneously type 1, 2, and 3. Such regions are called **simple solid regions**. The boundary of E is a closed surface, and we shall denote the positive orientation of the surface as outward (away from E), i.e. the unit normal vector $\mathbf{n}$ is directed outward.

ⓘ The Divergence Theorem

Let E be a simple solid region and let S be the boundary surface of E, given with positive (outward) orientation. Let $\mathbf{F}$ be a vector field whose component functions have continuous partial derivatives on an open region that contains E. Then

$$\iint_S \mathbf{F} \cdot \mathrm{d}\mathbf{S} = \iiint_E \operatorname{div} \mathbf{F} \cdot \mathrm{d}V$$

In other words, the flux of $\mathbf{F}$ across the boundary surface of E is equal to the triple integral of the divergence of $\mathbf{F}$ over E.

Too lazy to write a proof... so you'll have to imagine it (i.e., read the textbook for the proof).

Example of using the Divergence Theorem

Evaluate $\iint_S \mathbf{F} \cdot \mathrm{d}\mathbf{S}$, where

$$\mathbf{F}(x, y, z) = xy\mathbf{i} + (y^2 + e^{xz^2})\mathbf{j} + \sin(xy)\mathbf{k}$$

and S is the surface of the region E bounded by the parabolic cylinder $z = 1 - x^2$ and the planes $z = 0$, $y = 0$, and $y + z = 2$.

By the Divergence Theorem, $\iint_S \mathbf{F} \cdot \mathrm{d}\mathbf{S} = \iiint_E \operatorname{div} \mathbf{F} \cdot \mathrm{d}V$. Note that $\operatorname{div} \mathbf{F} = 3y$. And thus, considering the bounds, we have

$$\iint_S \mathbf{F} \cdot \mathrm{d}\mathbf{S} = \int_{-1}^{1} \int_0^{1-x^2} \int_0^{2-z} 3y \, \mathrm{d}y \, \mathrm{d}z \, \mathrm{d}x = \frac{184}{35}$$

Tip

The Divergence Theorem can be extended to regions that are finite unions of simple solid regions.

Regarding the above tip, consider, for example, a region E lying between closed surfaces S_1 and S_2 where S_1 is enclosed entirely by S_1. Let $\mathbf{n}_1$ and $\mathbf{n}_2$ be the positive (outward) normal vectors of S_1 and S_2, respectively. Then, the boundary surface of E is $S_1 \cap S_2$ and its normal $\mathbf{n}$ is given by $\mathbf{n} = -\mathbf{n}_1$ on S_1 and $\mathbf{n} = \mathbf{n}_2$ on S_2. Applying the Divergence Theorem gives us

$$\iiint_E \operatorname{div} \mathbf{F} \, \mathrm{d}v = -\iint_{S_1} \mathbf{F} \, \mathrm{d}\mathbf{S} + \iint_{S_2} \mathbf{F} \cdot \mathrm{d}\mathbf{S}$$

We can also apply he Divergence Theorem to fluid flow. Let $\mathbf{v}(x, y, z)$ be the velocity field with constant density ρ. Then $\mathbf{F} = p\mathbf{v}$ is the rate of flow per unit area, as defined previously. Let $P_0(x_0, y_0, z_0)$ be a point in the fluid and B_a be a ball with center P_0 and small radius a. Then $\operatorname{div} \mathbf{F}(P) = \operatorname{div} \mathbf{F}(P_0)$ for all other points P in B_a since $\operatorname{div} \mathbf{F}$ is continuous. Thus, the flux over the boundary sphere S_a can be approximated:

$$\iint_{S_a} \mathbf{F} \, \mathrm{d}\mathbf{S} = \iiint_{B_a} \operatorname{div} \mathbf{F} \, \mathrm{d}V = \iiint_{B_a} \operatorname{div} \mathbf{F}(P_0) \, \mathrm{d}V = \operatorname{div} \mathbf{F}(P_0) V(B_0)$$

Taking the limit as $a \to 0$, we derive

$$\operatorname{div} \mathbf{F}(P_0) = \lim_{a \to 0} \frac{1}{V(B_a)} \iint_{S_a} \mathbf{F} \cdot \mathrm{d}S$$

That is, $\operatorname{div} \mathbf{F}(P_0)$ is the net rate of outward flux per unit volume at P_0. In fact, this is the true motivation for the name *divergence*. If $\operatorname{div} \mathbf{F}(P) > 0$, the net flow near P is directed outward, and P is called a **source**. Conversely, if $\operatorname{div} \mathbf{F}(P) < 0$, the net flow near P is directly inward, and P is called a **sink**.

When reading a vector field, it's possible to estimate the divergence of $\mathbf{F}$ at a point P by observing the magnitude of the incoming and outgoing arrows. If the incoming arrows are longer than the outgoing arrows, P is a sink. Vice versa, and P is a source.

16.10 Summary

Fundamental Theorem of Line Integrals

$$\int_a^b F'(x)\,\mathrm{d}x = F(b) - F(a)$$

Fundamental Theorem for Line Integrals

$$\int_C \nabla f \cdot \,\mathrm{d}\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

Green's Theorem

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathrm{d}A = \int_C P\,\mathrm{d}x + Q\,\mathrm{d}y$$

Stokes' Theorem

$$\iint_S \mathrm{curl}\ \mathbf{F} \cdot \,\mathrm{d}\mathbf{S} = \int_C \mathbf{F} \cdot \,\mathrm{d}\mathbf{r}$$

Where $\mathrm{curl}\ \mathbf{F} = \nabla \times \mathbf{F}$.

Divergence Theorem

$$\iiint_E \mathrm{div}\ \mathbf{F} \cdot \,\mathrm{d}V = \iint_S \mathbf{F} \cdot \,\mathrm{d}\mathbf{S}$$

Where $\mathrm{div}\ \mathbf{F} = \nabla \cdot \mathbf{F}$.

Conclusion and Acknowledgements

Hope these notes were helpful and served as a more concise, understandable delivery of the content in the textbook! :)

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